

ORIGINAL ARTICLE

WILEY

Supermajority provisions and shareholders wealth: Evidence from South Korea's natural experiment

Hyeong Joon Kim  | Seung Hun Han 

School of Business and Technology
Management, College of Business, Korea
Advanced Institute of Science and Technology,
Daejeon, Republic of Korea

Correspondence

Seung Hun Han, School of Business and
Technology Management, College of Business,
Korea Advanced Institute of Science and
Technology, 291, Daehak-ro, Yuseong-gu,
Daejeon, Republic of Korea.
Email: synosia@kaist.ac.kr

Funding information

National Research Foundation of Korea,
Grant/Award Number: NRF-
2018H1A2A1060309

Abstract

Research Question/Issue: The existing research continues to debate whether firms' anti-takeover provisions enhance or harm shareholders' wealth. In this study, we examine the causal relation between the supermajority provision and shareholders' wealth by employing a quasi-natural experiment: the two court rulings that weaken the anti-takeover force of the supermajority provision in Korea, where the supermajority provision is the most widely used anti-takeover provision.

Research Findings/Insights: Using market reactions around the two court rulings, we find that firms with a supermajority provision as their only anti-takeover provision significantly underperform on average, compared with firms with no provision. This finding is robust to various empirical approaches that aim to address potential endogeneity concerns. We also find evidence that the supermajority provision plays a more significant role for firms with long-term investments, higher complexity, or higher takeover threats. Furthermore, the second among the two court rulings appears to have a stronger impact on the firm value and institutional investors' selling compared with the first one, suggesting the reinforcing effect of similar court decisions.

Theoretical/Academic Implications: Our study contributes to the corporate governance literature by investigating the effect of the supermajority provisions on shareholders' wealth. While conventional wisdom holds that the anti-takeover provisions harm shareholders' rights and wealth, our evidence from the quasi-natural experiment in Korea suggests that the positive effect of the supermajority provision empirically dominates the negative effect. Overall, our study supports the value-enhancing perspective on the supermajority provision, indicating that the Korean stock market views such a provision as inducing higher shareholders' wealth.

Practitioner/Policy Implications: This study highlights that a more contextual view of the relation between the anti-takeover provisions and shareholders' wealth is necessary. Moreover, we suggest that a regulation prohibiting firms' anti-takeover provisions might harm shareholders' wealth, but it appears that a firm-specific approach should be considered.

KEYWORDS

anti-takeover provisions, corporate governance, firm value, quasi-natural experiment, shareholders' wealth

1 | INTRODUCTION

Anti-takeover provisions remain one of the most controversial topics in the corporate governance literature. Conventional wisdom holds that such provisions weaken shareholders' rights and insulate directors from dismissal. Hence, under such provisions, incumbents may become inefficient at the shareholders' expense. However, by making removal harder, the anti-takeover provisions may allow directors and management sufficient time to create long-term shareholders' wealth. The literature has still not fully resolved this long-standing debate. In this study, we thus aim to shed light on these conflicting perspectives by focusing on the most widely used anti-takeover provision in our sample, under which firms require a supermajority-voting rule at their annual meetings.

The value-destroying perspective emphasizes the costs of anti-takeover provisions. Such provisions allow directors to become entrenched, even those who are self-interested, and thereby, agency problems arise. Theoretically, shirking, private benefit expropriation, and empire-building are more likely to be encouraged (Bertrand & Mullainathan, 2003; Jensen, 1993; Manne, 1965). In addition, Harris and Raviv (1988) show that rather than the simple majority voting rule, other majority rules (e.g., supermajority voting) that can be used as takeover defenses are socially suboptimal. As a result, the anti-takeover provisions increase agency costs (Jensen & Meckling, 1976) and, thus, are expected to reduce firm value. A stream of empirical studies supports this value-destroying perspective. In particular, Bebchuk et al. (2009) and Gompers et al. (2003) show the strong correlation between a higher number of anti-takeover provisions and lower firm value. Furthermore, several studies find that anti-takeover provisions (described as a lower quality of corporate governance) are negatively associated with overall performance, including accounting performance, reinvestment rates, beneficial acquisition offers, and post-merger performance (Bebchuk & Cohen, 2005; Cohen & Wang, 2013; Cremers & Ferrell, 2014; Faleye, 2007; Giroud & Mueller, 2011; Masulis et al., 2007).

The value-enhancing perspective argues that the anti-takeover provisions can be beneficial to the firms, as suggested in several theoretical studies (Chemmanur & Jiao, 2012; Stein, 1988, 1989). In particular, Stein (1988, 1989) posits that such provisions mitigate overinvestment in short-term projects and help directors avoid inefficient short-termism (or myopic pressure). Such provisions also can discourage directors from acting opportunistically toward the firm's stakeholders (e.g., large customers). In addition, the provisions reduce the stakeholders' risk by securing stability and continuity in management, as the firm's business plan cannot easily be reversed. Therefore, the cost of the relationship between a firm and its stakeholders decreases, and firm value improves (Knoeber, 1986; Shleifer & Summers, 1988).¹ In the case of takeover attempts, the anti-takeover provisions strengthen bargaining power and, thus, can extract a higher acquisition premium (DeAngelo & Rice, 1983; Stulz, 1988). Consistent with this value-enhancing view, empirical evidence supports the anti-takeover provisions raising shareholders' wealth (e.g., Cremers et al., 2017; Daines & Klausner, 2001). Specifically, a number of

studies show that the positive association between the anti-takeover provisions and firm value is significant for innovative firms; these studies support the theoretical argument related to myopic pressure (Bhojraj et al., 2017; Daines, 2001; Daines et al., 2018; Duru et al., 2013; Humphery-Jenner, 2014).

Although prior empirical studies robustly show a positive or negative relationship between the anti-takeover provisions and firm value, one might still argue that it is unclear whether such correlations are causal because of several endogeneity concerns.² In addition, there might be heterogeneity among the various anti-takeover provisions, making the estimation of the average effect of all the anti-takeover provisions difficult. Hence, it is worth focusing on an individual anti-takeover provision as the effect of the anti-takeover provisions is challenging to generalize. In this study, we aim to contribute to the large body of corporate governance literature through a quasi-natural experiment. Specifically, we exploit two court rulings that make the anti-takeover force of the supermajority provision potentially weaker for publicly traded firms in Korea, in line with prior research that used a legal context as a quasi-natural experiment (Bhargava et al., 2017; Cain et al., 2017; Cohen & Wang, 2013; Daines et al., 2018; Karpoff & Wittry, 2018; Larcker et al., 2011). We also note that our study estimates the effect of the supermajority provision (but not all the anti-takeover provisions) owing to the given nature of our empirical design.

In particular, we examine the cross-section of stock returns around the two court rulings in Korea in 2007 and 2008. We find a significantly negative market reaction over the 3-day window, -1.05 percentage points on average, for *treated firms* (i.e., firms with a supermajority provision as their only anti-takeover provision). Our baseline regressions also show that *treated firms* significantly underperform *control firms* (i.e., firms with no anti-takeover provisions). Overall, we find a negative association between shareholders' wealth and the two court rulings that weaken the anti-takeover force of the supermajority provision, consistent with *the value-enhancing perspective* (Bhojraj et al., 2017; Cremers et al., 2017; Daines et al., 2018; Daines & Klausner, 2001).

To further explore our main results in terms of the legal perspective, we investigate the impacts of the two court rulings (separately) including the data for trading behavior. Here, we find that the negative value impact of the second court ruling on *treated firms* is strongly supported, but that of the first court ruling is weakly supported. In addition, institutional investors are more likely to sell their shares of *treated firms* compared with *control firms*, particularly around the second court ruling. These results confirm the real effect of the precedent on the Korean stock market (although the Korean legal system follows the German-origin civil law). Taken together, we cautiously suggest that a reinforcing effect of the second court ruling, which has a similar court decision to the first court ruling, might be observed.

We adopt several empirical approaches to mitigate potential bias concerns. First, we use propensity score matching (PSM) as the market's views with respect to firm characteristics may differ between *treated* and *control firms*. Second, we conduct a sample of firms with anti-takeover provisions that did not change over the first and second

court rulings (i.e., the unchanged sample), as well as the changed sample. Third, we use two simulation approaches to assess the significance of our results. Finally, as robustness checks, we conduct a placebo test focusing on a sample of firms with one anti-takeover provision other than a supermajority provision, as their only anti-takeover provision, or use alternative fixed effects or alternative stock price reaction measures. Our main results continue to hold.

Next, we provide additional discussions. First, we incorporate large business group affiliations in Korea (i.e., the chaebol) and firm-specific governance quality scores into our analysis following the corporate governance literature that focuses on Korean samples (Baek et al., 2004; Baek et al., 2006; Black et al., 2006; Chang et al., 2017; Lee & Park, 2009; Shin & Park, 1999). The results indicate that our main findings are not driven by an omitted variable bias related to the chaebol group or the firm-level corporate governance quality. Second, we investigate the possible channels for the *value-enhancing perspective*. We find partial (but supportive) evidence that *treated firms* that engage in intensive R&D investments or operate in industries with higher complexity tend to experience a significant reduction in their firm value around the two court rulings. We also find that *treated firms* facing a higher takeover threat have a larger decrease in firm value. Collectively, the results of these additional tests support the argument that the supermajority provision plays a more significant role for subsets of firms.

Our study contributes to the corporate governance literature and provides policy implications. First, we construct our empirical setup by focusing on two court rulings that weaken the anti-takeover force of the supermajority provision; this is in line with prior studies using the legal setting as a natural experiment (e.g., Cohen & Wang, 2013). We present empirical evidence using several approaches, such as PSM and simulation. Second, we find that the changes in firm value around these two court rulings are, on average, consistent with the *value-enhancing perspective* of the supermajority provision. We perform an additional test to investigate the channels of our main findings. The literature argues that the market may view the anti-takeover provisions as having a heterogeneous effect across different subsets of firms (Amihud et al., 2017; Amihud & Stoyanov, 2017). Extending this line of thought, we find that the supermajority provision has a more significant effect on shareholders' wealth for firms with long-term investments, higher complexity, or higher threat of takeovers. Finally, our empirical evidence, in general, suggests that the importance of a more contextual view of the relation between the anti-takeover provisions and shareholders' wealth. For instance, for an industry where long-term investment is essential, the regulation made by policymakers who unconditionally regard the anti-takeover provisions as value-destroying corporate governance factors might harm shareholders' wealth.

The remainder of this paper is organized as follows. In Section 2, we discuss the relevant background to support the validation of our empirical framework. Section 3 describes our data and sample selection. In Section 4, we report the main results from the various empirical approaches. Section 5 further discusses our additional tests. Finally, Section 6 concludes the paper.

2 | EMPIRICAL FRAMEWORK

This section illustrates the relevant background of our empirical framework. In this study, we test whether weakening the anti-takeover force of a supermajority provision leads to a positive or negative effect on shareholders' wealth by using the two decisions from the local court in Korea (hereafter, the two court rulings), where the supermajority provisions are the most widely used anti-takeover provision. Thus, we begin by describing the institutional background in Korea and summarize the two court rulings and related anecdotal evidence.

2.1 | Institutional background in Korea

A company can have a provision in its bylaw or corporate charter that establishes higher voting requirements for an important action (e.g., approving a merger or replacing directors) compared with the threshold requirements in the law. This type of takeover defense is called the supermajority provision in the literature (Bebchuk et al., 2009; Gompers et al., 2003; Karpoff & Malatesta, 1989). The voting requirements in a supermajority provision typically exceed the participation level of shareholders at a meeting and, therefore, such important actions that require a supermajority rule become very costly or nearly unachievable (Field & Karpoff, 2002).

As our sample is composed of listed firms in Korea, we describe the three following points that make our empirical setup distinct. First, we mainly consider three anti-takeover provisions for our sample: supermajority provision, staggered board, and golden parachute. According to the Solidarity for Economic Reform (SER),³ public firms listed on the Korea Exchange (KRX) can usually adopt five types of (non-mutually exclusive) takeover defenses under the Commercial Act in Korea, as follows: (i) a supermajority rule for mergers, (ii) a supermajority rule for a change of directors, (iii) a supermajority rule for a change of charter, (iv) a staggered/classified board or a provision limiting the number of directors that can be replaced at a general meeting, and (v) a golden parachute.⁴ We consider takeover defense types (i)–(iii) to be the supermajority provision and type (iv) to be the staggered board. Overall, our focus is on these three anti-takeover provisions (typically, the supermajority provision).

Second, although focusing on the supermajority provision may contrast with the focus of prior studies, it is well suited to our empirical setup. In general, empirical studies on the value effect of corporate governance have focused on staggered boards in samples consisting of U.S. firms (Bebchuk & Cohen, 2005; Cohen & Wang, 2013; Cremers et al., 2017; Daines et al., 2018; Faleye, 2007). This is a reasonable approach because most U.S. firms have adopted staggered boards as takeover defenses (Bebchuk et al., 2002; Bebchuk & Cohen, 2005). Conversely, for Korean firms, the supermajority provision is the most widely used anti-takeover provision. According to the SER report mentioned above, Korean firms adopt takeover defense types (i)–(v) by about 1%, 14%, 8%, 3%, and 8%, respectively. The supermajority provision for a change of directors (i.e., type (ii)) is

the most selected anti-takeover provision by Korean-listed firms.⁵ This anecdotal evidence supports that focusing on the supermajority provision (rather than the staggered board or the golden parachute) is appropriate for our Korean sample.

Third, in contrast with U.S. firms, all Korean firms are affected by a single law, the Commercial Act in Korea (i.e., there are no state or local laws in Korea).⁶ We also note that the corporate charter in Korea is conceptually the same as combining both corporate charter and bylaws in United States. For instance, matters that can be included in bylaws for U.S. firms should be included in charters for Korean firms.⁷ Therefore, we simply use the term “charter” (*Jeong-Gwan* in Korean) hereafter. We collect the charters of firms in our data collection process (naturally, bylaws are not considered) to construct the variables of the anti-takeover provisions. We provide detailed definitions of our variables in Section 3.

Meanwhile, our study can only be valuable to the corporate governance literature if there is an active takeover market in South Korea, even though several studies provide insights into corporate governance using non-U.S. samples. Similar to Frattaroli (2020), we compare Korea with the G7 countries (Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States) and China in terms of the mergers and acquisitions (M&A) market in the 2000s. The statistics show that average annual number and total volume of M&A in the Korean market are sufficiently significant (for brevity, these are reported in the Online Appendix). Overall, Korea has an active market for M&A in the 2000s, indicating that our study can be of interest to the corporate governance literature.

2.2 | The two court rulings related to the anti-takeover force of the supermajority provision

In this subsection, we discuss the two court rulings related to the supermajority provision in the charter of a firm. The first and second court rulings were made by local courts in Incheon and Seoul in 2007 and 2008, respectively (Incheon metropolitan city is the third most populous city in South Korea). We then discuss the Korean commercial legal tradition and provide anecdotal evidence related to the two court rulings.

2.2.1 | The first court ruling

On February 20, 2007, the respondent firm of the first court ruling, FYD Corp. (later Provita Corp.), had changed its charter to impose a supermajority provision for the removal of a director at the general shareholders' meeting. On March 6, 2007, minority shareholders of FYD turned to the local court seeking to invalidate the general shareholders' meeting (held on February 20, 2007). They especially argued that the resolution adopted of “a supermajority provision for the dismissal of a director” infringed their own rights as shareholders. On April 13, 2007, the Incheon District Court partially approved the provisional disposition, suspending the validity of the resolutions of the general shareholders' meeting raised by minority shareholders. In short, the Incheon District Court (partially) agreed with minority

shareholders and invalidated the supermajority provision of FYD. Three judges of the Incheon District Court made the following declaration: Incheon District Court (Bucheon Branch) Order 2007Kahap335 dated April 13, 2007 [Provisional Disposition Suspending Validity of Resolution of General Shareholders' Meeting].

... It cannot be allowed to impair the essence of the corporation, hamper the purpose of its existence, or infringe on the inherent rights of the shareholders. ... The supermajority provision in the charter of the respondent firm can result in the serious distortion of shareholder voting rights, therefore such a provision will be invalid. ...

2.2.2 | The second court ruling

On March 28, 2008, the respondent firm of the second court ruling, GNCO Co., Ltd., held a general shareholders' meeting. In particular, GNCO changed its directors and revised its charter to adopt a supermajority provision for the removal of a director. On April 8, 2008, a minority shareholder of GNCO (a trustee of the fund in which approximately 8% of GNCO's shares are incorporated) and a former director, who was dismissed at the last shareholders' meeting, argued that there were several shortcomings in the last meeting and turned to the local court, seeking to invalidate the meeting. On June 2, 2008, the Seoul Central District Court partially approved the provisional disposition suspending the validity of resolutions of the general shareholders' meeting. In short, the Seoul Central Court (partially) agreed with minority shareholders and, thus, invalidated the supermajority provision of GNCO. Three judges of the Seoul Central District Court made the following declaration, which was similar to the first court ruling: Seoul Central District Court Order 2008Kahap1167 dated June 2, 2008 [Provisional Disposition Suspending Validity of Resolution of General Shareholders' Meeting, etc.].

... When the directors are improperly managing, shareholders need to be able to quickly dismiss directors in order to protect their property. For this purpose, the Commercial Act clarifies that, at any time, the directors can be removed by the special-decision in the shareholders' meeting.⁸ ... The voting requirements which are stricter than the special-decision set in the Commercial Act could result in suppressing a large number of shareholders by specific shareholders or giving veto power to specific shareholders. Therefore, such a supermajority provision in the charter of the respondent firm contradicts the Commercial Act. Hence, adopting such a provision is an invalid act. ...

2.2.3 | Legal force of the court rulings in Korea

One might argue about the effect of the court rulings in Korea on the stock market, since the Korean commercial legal tradition follows

the German civil law system (La Porta et al., 1997, 1998, 1999). Therefore, we note two important arguments on the effect of the court rulings due to the nature of the difference between civil and common law systems. First, if the precedent (in Korea) has a strong binding force similar to the common law system (i.e., the doctrine of *stare decisis*), then we expect that participants in the Korean market also react similarly to ones in the common law system. In particular, as the first court ruling already releases information by casting a doubt on the validity of the supermajority provision, no significant market reaction should be observed around the second court ruling (or any similar ruling after the first court ruling). Second, if the precedent has no actual legal force in any way, then we should never observe an abnormal market reaction around the two court rulings (also, all the court rulings related to the corporation). However, the reality could lie between these two arguments given that the Korean law system only concedes the “*de facto*” legal binding force of the precedent. We focus on this view in Section 4.3.

2.2.4 | Anecdotal evidence

In October 2007, *Joong-Ang Daily*, a major media company in Korea, reported that “the supermajority provision to defense (sic) merger and acquisition is invalid” by using the declaration of the first court ruling.⁹ Meanwhile, *The Law Times*, the sole legal newspaper in Korea, reports the second court ruling in full in June 2008.¹⁰ Moreover, the Korea Corporate Governance Services (KCGS)¹¹ mention that, by citing the second court ruling, “the supermajority-voting rule is arguable, since recently there was a court decision that the supermajority which is stricter than the threshold described in the Commercial Act, is contrary to the purpose of the Commercial Act.”¹²

The respondent firm of the second court ruling, GNCO Co., Ltd., which was listed on the Korean Securities Dealers Automated Quotation (KOSDAQ) in KRX, experienced a stock price decrease of about 5.1% within a post-trading day of the court ruling (i.e., from the close of market on June 2 to the close of market on June 3, 2008).¹³ This market reaction was an economically significant decrease compared with the indexes of KOSDAQ and Korea Composite Stock Price Index (KOSPI) over the same period, which decreased by 1.1% and 1.5%, respectively. Therefore, the respondent firm was clearly (and abnormally) affected by its court ruling (i.e., the invalidation of its supermajority provision), which reduced the firm value. In other words, market participants may have considered the supermajority provision of the respondent firm as raising shareholders' wealth.¹⁴

Consistent with the anecdotal evidence, market participants may perceive that a firm's supermajority is not as robust due to the two court rulings. For example, a firm's minority shareholders can nullify the supermajority provision by taking similar legal action to those associated with the two court rulings. As a result, the anti-takeover force of the supermajority provision may have been weakened by these court rulings. This is where our empirical setup becomes worthwhile.¹⁵ Overall, we argue that the supermajority provision of a firm is more likely to be invalid in the legal context. We also argue that

participants in the Korean financial market are aware of the possibility of the supermajority provision becoming invalid.

Nevertheless, we conduct a placebo test to further support the validity of our empirical framework. Focusing on the firms that have a staggered board or golden parachute as their only anti-takeover provision, wherein these firms are expected to be not affected by the two court rulings in any way, we find that stock price reactions of such firms are not significantly different from those of firms with no provisions. The results of the placebo test are provided in the Online Appendix. We also discuss whether participants in the Korean market, such as institutional investors, abnormally react to the most affected firms by the court rulings (i.e., firms with the supermajority provisions) in Section 4.3.

2.3 | The natural experiment based on announcements of the two court rulings

We use a quasi-natural experiment setup based on the two court rulings that make the anti-takeover force of the supermajority provision potentially weaker. In particular, we follow Cohen and Wang (2013) to investigate whether the two court rulings led to an increase or a decrease in affected firms' shareholders' wealth. Cohen and Wang (2013) focus on two court rulings at the Delaware Chancery Court in October 2010 and the Delaware Supreme Court in November 2010 as a natural experiment setup. The former court ruling weakened the potential anti-takeover force of staggered boards, whereas the subsequent ruling by the Delaware Supreme Court reversed that decision. Specifically, Cohen and Wang (2013) integrate the two events by pooling the observations for abnormal returns and conclude that staggered boards are negatively associated with firm value using the returns at the time the two Delaware court rulings were announced.

Extending the view of Cohen and Wang (2013), we argue that our empirical design can estimate the underlying effect on firm value of the decisions that weakened the supermajority provision. Collectively, we examine the cross-section of stock returns around the announcement of the two court rulings and estimate the wealth effect for affected firms (i.e., firms that have a supermajority provision prior to the two court rulings). We also pool the two court rulings in our empirical analysis to take advantage of the increased sample size and power of the test (Cohen & Wang, 2013).¹⁶ The prediction based on the two conflicting perspectives of the anti-takeover provisions in the literature is summarized below.

If market participants view the firm's supermajority provision as a value-destroying (i.e., if the *value-destroying perspective* is supported), then weakening of the anti-takeover force of firms with the supermajority provision should be positively associated with firm value on average. Consequently, abnormal returns for the impacted firms that have a supermajority provision should be higher than those for non-impacted ones at the time of the two court rulings. On the contrary, if the supermajority provision is regarded as a value-enhancing tool (i.e., if the *value-enhancing perspective* is supported), the impacted

firms should underperform compared with non-impacted firms around the time of the two court rulings.

3 | DATA AND SAMPLE SELECTION

3.1 | Data

We first obtain data for our main independent variables, the anti-takeover provisions, in the charters of firms listed on the KRX from the Data Analysis, Retrieval and Transfer System (DART).¹⁷ DART is provided by the Financial Supervisory Service, the independent agency under the Financial Services Commission of the Korean Government, which is similar to the U.S. Securities and Exchange Commission (SEC). We exclude firms in the financial and utilities industries. To identify the exact statement of the firm's charter, we hand-collect charter data from DART immediately before the announcement dates of the two court rulings. We also obtain annual financial statement information from FN Dataguide and board information from TS2000. These data sets are similar to Compustat and Execucomp, respectively, for U.S. firms.

We measure the market reaction in terms of the cumulative abnormal returns (CARs) from event day -1 to event day $+1$ using the market model ($CAR[-1, +1]$). The event day is the date of the announcement of the two court rulings: April 13, 2007, for the first court ruling and June 2, 2008, for the second court ruling. The parameters of the market model are estimated using the past 180 trading days. We use the KOSPI and KOSDAQ value-weighted indexes as the market return parameters for firms listed on the KOSPI and KOSDAQ, respectively. We obtain the data on stock prices and market returns from FN Dataguide.

To accurately estimate the effects of the two court ruling events, we need to exclude samples that could result in spurious treatment effects. If the court ruling on which we focus was not exogenous for the respondent firm, then its abnormal returns following the ruling may not solely reflect the impact of invalidation of its supermajority provision (Karpoff & Wittry, 2018). Thus, we exclude the respondent firms (as described in Section 2.2) for each

court ruling. Moreover, we check whether a firm has confounding events during our data-collection process because several firm-specific events around the two court rulings may influence the market reaction on the event dates.¹⁸ We exclude observations that have such events 2 weeks before and after the dates of the two court rulings.¹⁹ For our final sample, we exclude observations in the top and bottom one percentile of the stock price reaction distribution so that outliers do not drive our results (Graham et al., 2003). Table 1 reports the summary of our final sample of 2,286 observations available to calculate the stock price reaction around the two court rulings.

3.2 | Summary statistics of the sample

We classify the observations in Table 1 into the three main types of anti-takeover provisions and one miscellaneous category: *Supermajority*, *Staggered Board (SB)*, *Golden Parachute (GP)*, and *Miscellaneous Provisions (Miscell)*.²⁰ Consistent with the anecdotal evidence in Section 2.1, Table 1 shows that a supermajority provision is the most adopted anti-takeover provision in our final sample.

As our key independent variable, *Supermajority*, we consider that a firm has a supermajority provision if its charter requires a voting rate above the special-decision threshold stated in the Commercial Act in Korea. Specifically, we consider whether the firm has a supermajority provision for the following three resolutions: (1) approving a merger, (2) changing directors, and (3) changing the provisions in a charter. As these resolutions are directly related to takeovers, the Commercial Act requires that they be determined via a special-decision in the shareholders' meeting (rather than by a general-decision), for every firm in Korea. We focus on the firms with supermajority provisions for these important resolutions. In our sample, such firms typically require 90% of the voting rights of the shareholders attending a general meeting and 70% of the total issued and outstanding shares (compared with 66.67% and 33.33%, respectively, in the Commercial Act in Korea). In other words, it is virtually infeasible to change the directors of such firms.

TABLE 1 Sample summary

	Supermajority	Staggered board	Golden parachute	Miscellaneous provisions	At least one anti-takeover provision	Only supermajority (Treated firms)	Zero anti-takeover provisions (Control firms)
Event1 (N = 1,088)	117 (10.75%)	28 (2.57%)	51 (4.69%)	7 (0.64%)	160 (14.71%)	82 (7.54%)	928 (85.29%)
Event2 (N = 1,198)	159 (13.27%)	36 (3.01%)	74 (6.18%)	6 (0.50%)	200 (16.69%)	95 (7.93%)	998 (83.31%)
Total (N = 2,286)	276 (12.07%)	64 (2.80%)	125 (5.47%)	13 (0.57%)	360 (15.75%)	177 (7.74%)	1,926 (84.25%)

Note: This table presents the sample summary. Our sample consists of non-financial firms listed on KRX for the two court ruling events. The first court ruling date was April 13, 2007, and the second one was June 2, 2008. *Event1* and *Event2* indicate the first and second court ruling events, respectively. We manually check the anti-takeover provisions in the charter of the firm before the two court ruling events, using firms' annual reports available in the Data Analysis, Retrieval and Transfer System (DART). We exclude samples that had other events 2 weeks before and after the court ruling announcement dates. We classify the firm-year observations into several types of anti-takeover provisions and report the number of samples and proportions as a percentage in parentheses. Before the two court rulings, *treated firms* have a supermajority provision as their only anti-takeover provision, and *control firms* are firms having no anti-takeover provisions. A detailed explanation of the variables is in Appendix A. *N* denotes the number of observations.

Furthermore, we note that the Commercial Act in Korea does not enforce the minimum requirement to attend the shareholders' meeting for either special or general-decisions. For instance, in the case where one shareholder presents at the meeting, he or she can adopt a general-decision resolution only if he or she holds more than 25% of voting rights. If he or she holds more than 33% of voting rights, he or she can adopt any resolution (either special or general-decisions). Therefore, we consider that a firm has a supermajority provision if it has a minimum requirement in terms of voting rights for the shareholders attending a general meeting to adopt the three resolutions outlined above. Such firms in our sample typically require that at least 50% of the total issued and outstanding shares should attend a shareholders' meeting (otherwise, no resolution can be adopted), which again is virtually impossible to achieve when a takeover situation arises. For example, if the majority owner does not agree on a resolution related to a takeover, then he or she can prevent it by being absent from the meeting.

In addition, we consider that a firm has a *staggered board* for firms not only with divided boards (usually three) but also with a provision limiting the number of directors (in our sample, typically one-third or one-quarter) that can be replaced at a shareholders' meeting. For example, if a firm has a provision in its charter that only one-third of the board of directors can be replaced at the

general meeting, then we consider such a firm the same as one with a three-staggered board structure.

Table 2 reports the descriptive statistics of our final sample. To avoid the potential impact of outliers, firm characteristics (which are continuous variables) are winsorized at the 1st and 99th levels. All the variables from the annual statements of firms are measured in the fiscal year prior to the year of the court ruling announcements. Appendix A provides detailed definitions of the variables.

4 | MAIN RESULTS

4.1 | Market reactions around the two court ruling events

In our empirical setting, firms with a supermajority provision are expected to be affected by the two court rulings, whereas firms with no anti-takeover provision (*control firms*) have no anti-takeover force to be affected. However, although the two court rulings in our study weaken the anti-takeover force of the supermajority provision, the firm with both a supermajority provision and a staggered board may be affected differently (it is likely that the impact is less for these firms than for the firms with only a supermajority provision). Therefore, our

TABLE 2 Descriptive statistics

Variables	N	Mean	SD	p25	Median	p75
CAR[−1, 1] (%)	2,286	0.535	5.087	−2.556	0.100	2.784
Supermajority	2,286	0.121	0.326	0.000	0.000	0.000
Staggered board (SB)	2,286	0.028	0.165	0.000	0.000	0.000
Golden parachute (GP)	2,286	0.055	0.227	0.000	0.000	0.000
Miscellaneous provisions (Miscell)	2,286	0.006	0.075	0.000	0.000	0.000
Other provisions	2,286	0.080	0.271	0.000	0.000	0.000
Age	2,239	27.717	15.333	14.000	26.000	38.000
Size	2,239	18.582	1.339	17.641	18.305	19.248
Cash	2,239	0.077	0.078	0.021	0.052	0.105
Leverage	2,239	0.432	0.207	0.263	0.433	0.591
M-to-B	2,239	1.649	1.849	0.707	1.102	1.772
OCF	2,239	0.034	0.111	−0.017	0.041	0.100
ROA	2,239	0.033	0.098	−0.000	0.040	0.088
Investment	2,239	0.062	0.076	0.017	0.045	0.092
Stock volatility	2,239	0.029	0.016	0.018	0.025	0.036
Stock run-up	2,239	0.055	0.140	−0.026	0.044	0.121
Board size	2,239	1.606	0.347	1.386	1.386	1.946
Board independence	2,239	0.250	0.159	0.250	0.250	0.333
Controlling shareholder ownership	2,239	0.399	0.168	0.280	0.390	0.513

Note: This table presents the descriptive statistics of our sample. We manually check the anti-takeover provisions in the charter of the firm before the two court ruling events, using firms' annual reports available in the Data Analysis, Retrieval and Transfer System (DART). We exclude samples that had other events 2 weeks before and after the court ruling announcement dates. The variables from annual statements are measured in the fiscal year before the court rulings. We winsorize the continuous variables at the 1st and 99th percentiles. A detailed explanation of the variables is in Appendix A. N, SD, p25, and p75 denote the number of observations, standard deviations, and 25th and 75th percentiles, respectively.

analysis basically focuses on the firms with a supermajority provision as their sole anti-takeover provision to defend their management rights (*treated firms*). Specifically, we define the variable, *Treated*, as one for *treated firms* and zero for *control firms*.²¹

Panel A of Table 3 reports the univariate analysis for CARs around the two court rulings. Consistent with the market viewing the

supermajority provision as a value-enhancing tool, the mean (median) of announcement returns are on average -1.05% (-1.36%) for *treated firms* at the 1% level.²² The differences in both mean and median between *treated* and *control firms* are all significantly negative at the 1% level. In other words, we find that weakening the anti-takeover force of supermajority provision by the two court rulings

TABLE 3 Market reactions around the court rulings

Panel A: Univariate test			
CAR[−1, +1]	<i>Treated firms</i> (treated = 1) (1)	<i>Control firms</i> (treated = 0) (2)	Difference (3)
Mean (t statistics)	−1.05%*** (2.74)	0.67%*** (5.90)	−1.72%*** (4.40)
Median (z statistics)	−1.36%*** (3.49)	0.18%*** (3.38)	−1.54%*** (4.42)
N	177	1,926	
Panel B: Baseline regression			
Dependent variable = CAR[−1, +1] (%)			
	(1)	(2)	(3)
Treated	−1.684*** (0.403)	−1.431*** (0.389)	−1.520*** (0.407)
Event2	0.029 (0.210)	−0.033 (0.216)	−0.039 (0.216)
Age		0.019** (0.009)	0.010 (0.009)
Size		0.150 (0.118)	0.214* (0.128)
Cash		0.888 (1.744)	1.449 (1.723)
Leverage		0.375 (0.648)	0.075 (0.682)
M-to-B		0.043 (0.095)	0.075 (0.101)
OCF		−0.735 (1.405)	−0.827 (1.470)
ROA		0.103 (1.671)	0.496 (1.706)
Investment		0.240 (1.620)	0.029 (1.649)
Stock volatility		−27.687*** (9.004)	−28.706*** (9.229)
Stock run-up		0.908 (1.101)	0.483 (1.098)
Board size		−0.464 (0.386)	−0.322 (0.392)
Board independence		0.382 (0.908)	−0.062 (0.933)
Controlling shareholder ownership		−0.121 (0.752)	−0.679 (0.776)
Industry fixed effect	Yes	No	Yes
N	2,103	2,060	2,060
Adj. R ²	2.39%	1.70%	3.00%

Note: This table presents the results of average stock price reaction around the two court rulings and baseline OLS regressions. Cumulative abnormal returns (CAR[−1, +1]) are measured from a day before the announcement date to a day after the announcement date using the market model estimated over the past 180 trading days. The first court ruling date was April 13, 2007, and the second one was June 2, 2008. Before the two court rulings, *treated firms* have a supermajority provision as their only anti-takeover provision, and *control firms* have no anti-takeover provisions. In panel A, we use *t* tests and Wilcoxon tests for the means and medians, respectively. All tests are based on the null hypothesis that the value or the difference is equal to zero. *N* denotes the number of observations. In panel B, the industry fixed effect is based on the 2-digit KSIC dummies, and the intercept is included. A detailed explanation of the variables is in Appendix A. Robust standard errors adjusted for heteroscedasticity and clustered by firm-level are reported in parentheses.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

harms shareholders' wealth. Panel B of Table 3 reports our baseline regression for the differences in market reactions at the time of the two court rulings. To account for the possible differences in the mean of announcement returns between the two court ruling events, we include an event dummy (*Event2*). We also include several firm characteristics as follows: firm age (*Age*), total assets (*Size*), the ratio of cash to total assets (*Cash*), the ratio of debt to total assets (*Leverage*), the market-to-book ratio (*M-to-B*), the ratio of operating cash flow to total assets (*OCF*), return on assets (*ROA*), the ratio of total expenses on research and development and capital expenditure to total assets (*Investment*), the number of directors (*Board size*), the proportion of independent directors (*Board independence*), and the proportion of shareholdings held by the controlling shareholder (*Controlling shareholder ownership*). In addition, we consider the following pre-event control variables related to abnormal returns: 1-month cumulative stock volatility prior to the announcement (*Stock volatility*), and 1-month cumulative stock returns prior to the announcement (*Stock run-up*). To control the industry fixed effect, we include industry dummies based on the first two-digit codes of the Korean Standard Industrial Classification (KSIC).

Table 3 overall shows that *treated firms* significantly underperform compared with *control firms* by about 1.43–1.52 percentage points over the 3-day event window after including firm characteristics as the control variables. Therefore, the estimation results in Table 3 suggest that weakening the anti-takeover force of the supermajority provision is negatively associated with firm value. In other words, market participants view the supermajority provision as a tool for enhancing shareholders' wealth.

Note that our primary analysis classifies the observations into three groups: *treated*, *control firms*, and omitted observations. These omitted observations have multiple provisions or another provision (other than a supermajority provision) as the only one anti-takeover provision, as the anti-takeover provisions are not mutually exclusive in the firm. Thus, we perform regressions with indicators of each anti-takeover provision and its interaction effects for the full sample (Table 4) that has all the possible combinations of anti-takeover provisions. Specifically, we include *Supermajority*, *SB*, *GP*, and *Miscell* as well as the interaction terms (*Supermajority* × *SB* and *Supermajority* × *GP*) in columns (1)–(4), whereas the anti-takeover provisions other than the supermajority provision (an indicator, *Other-provision*) and its

TABLE 4 Multivariate regression for the anti-takeover provisions

Dependent variable = CAR[−1, +1] (%)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Supermajority	−1.266*** (0.380)	−1.730*** (0.394)	−1.124*** (0.386)	−1.567*** (0.396)	−1.028*** (0.375)	−1.681*** (0.401)	−0.896** (0.384)	−1.522*** (0.403)
SB	1.377* (0.745)	−0.824 (0.889)	1.054 (0.812)	−1.194 (1.031)				
GP	0.897 (0.650)	−0.327 (1.018)	1.213* (0.676)	0.053 (1.064)				
Miscell	−2.333** (1.026)	−2.370** (1.020)	−2.435** (1.092)	−2.483** (1.080)				
Supermajority × SB		3.576*** (1.259)		3.602*** (1.355)				
Supermajority × GP		2.014 (1.281)		1.915 (1.306)				
Other-provision					0.569 (0.511)	−0.778 (0.662)	0.661 (0.537)	−0.671 (0.706)
Supermajority × Other-provision						3.057*** (0.969)		2.983*** (0.991)
Event2	−0.019 (0.208)	−0.034 (0.207)	−0.0429 (0.214)	−0.061 (0.213)	−0.015 (0.208)	−0.040 (0.208)	−0.039 (0.214)	−0.063 (0.214)
Control variables	No	No	Yes	Yes	No	No	Yes	Yes
Industry fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2,286	2,286	2,239	2,239	2,286	2,286	2,239	2,239
Adj. R ²	2.69%	3.10%	3.27%	3.65%	2.39%	2.90%	2.97%	3.45%

Note: This table presents the results from OLS regressions where the dependent variables are the CARs around the two court rulings. Control variables are the same as in Table 3. A detailed explanation of the variables is in Appendix A. Industry fixed effect is based on the 2-digit KSIC dummies, and the intercept is included. Robust standard errors adjusted for heteroscedasticity and clustered by firm-level are reported in parentheses.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

TABLE 5 Propensity score matching results

Panel A: Propensity score estimation		
Dependent variable = treated		
	Pooled (1)	PSM (2)
Age	0.002 (0.004)	0.000 (0.006)
Size	−0.174*** (0.063)	0.086 (0.097)
Cash	−0.273 (0.632)	−0.546 (0.934)
Leverage	0.114 (0.274)	−0.145 (0.445)
M-to-B	−0.019 (0.023)	0.050 (0.046)
OCF	0.334 (0.503)	−0.431 (0.788)
ROA	−0.504 (0.570)	1.047 (0.925)
Investment	0.438 (0.637)	−0.204 (1.024)
Stock volatility	3.696 (3.236)	2.623 (4.823)
Stock run-up	−0.217 (0.327)	0.063 (0.516)
Board size	0.320* (0.173)	−0.094 (0.267)
Board independence	−0.317 (0.371)	0.061 (0.516)
Controlling shareholder ownership	−1.311*** (0.405)	0.320 (0.550)
Constant	1.704 (1.123)	−1.585 (1.587)
N	2,060	328
Pseudo R^2	5.66%	1.25%
Panel B: Sample balance		
Differences in mean		
	Pooled (1)	PSM (2)
Propensity score	0.038***	−0.009
Age	−2.689**	1.425
Size	−0.512***	0.155
Cash	0.004	−0.008
Leverage	−0.008	0.010
M-to-B	0.256*	0.092
OCF	−0.017**	0.002
ROA	−0.028***	0.014
Investment	0.005	−0.003
Stock volatility	0.004***	0.001
Stock run-up	−0.010	0.006
Board size	−0.001	0.004
Board independence	−0.028**	0.007
Controlling shareholder ownership	−0.080***	0.020
Panel C: Matched sample regression		
Dependent variable = CAR[−1, +1] (%)		
	(1)	(2)
Treated	−1.745*** (0.635)	−1.788*** (0.645)
Event2	−1.083* (0.643)	−1.401** (0.668)
Control variables	No	Yes
Industry fixed effect	Yes	Yes

(Continues)

TABLE 5 (Continued)

Panel C: Matched sample regression		
Dependent variable = CAR[−1, +1] (%)		
	(1)	(2)
N	328	328
Adj. R ²	4.94%	3.96%

Note: This table presents the results from the propensity score matched (PSM) sample. PSM sample is conducted using the propensity score estimated in panel A as follows: one-to-one nearest neighborhood matching with non-replacement, matching within the same event (the first and second court ruling) and same listing venue (KOSPI and KOSDAQ), and dropping 5% of the treatment observations for which the propensity score density of the matched observations is the lowest. Panel A reports the results of propensity score estimation by the probit model where the dependent variable is *Treated*. Panel B reports the differences in means of the propensity score (estimated in panel A) and firm characteristics. Panel C reports the regression results using the matched sample. A detailed explanation of the variables is in Appendix A. Industry fixed effect is based on the 2-digit KSIC dummies, and the intercept is included. Robust standard errors adjusted for heteroscedasticity and clustered by firm-level are reported in parentheses.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

interaction with *Supermajority* in columns (5)–(8). Consistent with our previous finding, all the coefficients of *Supermajority* are significantly negative at the 1% level. Furthermore, the interaction terms have positive coefficients, suggesting that firms with multiple anti-takeover provisions (including a supermajority provision) are less affected than *treated firms* around the court rulings. It is reasonable the two court rulings are specifically related to the validity of a supermajority provision.

4.2 | Propensity score matching

Next, to mitigate the potential endogeneity concern, we use the PSM method, which is widely employed in the literature to examine causality (Heckman et al., 1997, 1998; Rosenbaum & Rubin, 1983). There are concerns about selection bias because the treatment group in our study (i.e., *treated firms* having a supermajority provision as the only anti-takeover provision) might not be randomly assigned. In other words, the differences in announcement returns between *treated* and *control firms* may reflect the reaction of the market with respect to certain firm characteristics, rather than the causal relation between the weakening of the supermajority provision as an anti-takeover provision and shareholders' wealth. Therefore, in line with Cohen and Wang (2013) and Cremers et al. (2009), we compare *treated* and *control firms* that have similar firm characteristics.

In panel A of Table 5, we estimate the likelihood of being a *treated firm*. We use the probit model to calculate the propensity score (Rosenbaum & Rubin, 1983; Smith & Todd, 2005). Three characteristics (*Size*, *Board size*, and *Controlling shareholder ownership*) are significantly related to the likelihood of being a *treated firm*. In other words, it is possible that the market reacts differently to certain firm characteristics (e.g., negatively to smaller firms) rather than to the weakening of the anti-takeover force of the supermajority provision. Therefore, the matching approach can be an appropriate alternative to address this potential bias concern.

We construct a matched sample (hereafter, the PSM sample) by using one-to-one nearest neighborhood matching with non-replacement. In addition, to improve our matching quality, we match the sample within the same event (i.e., the same fiscal year) and the listing venue (KOSPI vs. KOSDAQ).²³ Furthermore, we drop 5% of the treatment observations for which the propensity score density of the control observations is lowest, as suggested by Smith and Todd (2005). We repeat the probit regression for the PSM sample in column (2) of panel A of Table 5. The insignificant coefficients for all the variables show that our PSM sample is well balanced on average with respect to all the matching variables. Moreover, we test the differences in the means of the firm characteristics between *treated* and *control firms*. In panel B of Table 5, columns (1) and (2) report the results for our unmatched and PSM sample, respectively. Overall, our PSM sample shows a good balance in all the covariates and estimated propensity scores.

Panel C of Table 5 reports the ordinary least squares (OLS) regression results for our PSM sample. In both columns, the estimated coefficients of *Treated* remain significantly negative. Consistent with the argument that market participants view the supermajority provision as a value-enhancing tool, *treated firms* significantly underperform around the two court rulings by about 1.74–1.79 percentage points relative to *control firms* that have similar firm characteristics.

4.3 | Reinforcing effect of the precedent and trading behavior around the court rulings

Our results suggest that (not only the amendment of the Commercial Act in Korea but also) the court rulings related to the corporation can significantly affect the stock market. In other words, although Korea is one of the civil law countries, the precedent in Korea appears to have some actual legal force (or “de facto” legal binding force). Nevertheless, the different impacts of the two court rulings are worthy to

TABLE 6 Reinforcing effect of the precedent: trading behavior around the first and second court rulings

Panel A: Correlation matrix						
	(1)	(2)	(3)	(4)	(5)	(6)
Institutional investors NB[−1, +1]: (1)	1.00					
Individual NB[−1, +1]: (2)	−0.54***	1.00				
Foreign investors NB[−1, +1]: (3)	−0.32***	−0.30***	1.00			
Total trading volume[−1, +1]: (4)	−0.03	0.04**	−0.01	1.00		
CAR[−1, +1]: (5)	0.06***	−0.05**	0.01	0.12***	1.00	
Treated: (6)	−0.02	0.03	−0.01	0.06***	−0.10***	1.00
Panel B: Institutional investors' trading pattern						
Dependent variable = Institutional investors NB[−1, +1] (%)						
	Pooled (1)	PSM (2)	First court ruling (Event2 = 0) (3)	First-PSM (Event2 = 0) (4)	Second court ruling (Event2 = 1) (5)	Second-PSM (Event2 = 1) (6)
Treated	−0.501 (0.673)	−1.047 (0.939)	0.979 (1.144)	−0.727 (1.594)	−1.683** (0.830)	−2.660* (1.374)
Event2	0.336 (0.542)	−1.436 (1.126)				
N	2,046	326	985	138	1,061	188
Adj. R ²	−0.18%	8.30%	3.42%	25.30%	−1.19%	10.30%
Panel C: Trading volume around the two court rulings						
Dependent variable = Total trading volume[−1, +1] (%)						
	Pooled (1)	PSM (2)	First court ruling (Event2 = 0) (3)	First-PSM (Event2 = 0) (4)	Second court ruling (Event2 = 1) (5)	Second-PSM (Event2 = 1) (6)
Treated	−0.382 (0.997)	0.277 (1.524)	0.274 (1.847)	−0.832 (3.755)	−1.159 (0.993)	−0.054 (0.877)
Event2	−4.044*** (0.451)	−5.997*** (1.337)				
N	2,046	326	985	138	1,061	188
Adj. R ²	25.30%	24.20%	27.70%	18.90%	12.80%	8.45%
Panel D: Relationship between CARs and institutional investors' trading pattern						
Dependent variable = CAR[−1, +1] (%)						
	Pooled (1)	(2)	First court ruling (Event2 = 0) (3)	(4)	Second court ruling (Event2 = 1) (5)	(6)
Institutional investors NB[−1, +1]	0.020*** (0.007)	0.020*** (0.007)	0.024* (0.013)	0.024* (0.013)	0.023** (0.009)	0.024*** (0.009)
Institutional investors NB[−1, +1] × Treated	0.037 (0.055)	0.036 (0.056)	0.021 (0.074)	0.018 (0.077)	0.065* (0.036)	0.074* (0.038)
Treated	−1.542*** (0.409)	−1.254*** (0.471)	−0.783 (0.679)	−0.889 (0.782)	−2.092*** (0.517)	−1.116** (0.499)
Total trading volume [−1, +1]		0.099*** (0.016)		0.118*** (0.019)		0.092*** (0.027)
Total trading volume [−1, +1] × Treated		−0.034 (0.062)		0.007 (0.061)		−0.249*** (0.031)
Event2	−0.052 (0.217)	0.329 (0.222)				

(Continues)

TABLE 6 (Continued)

Panel D: Relationship between CARs and institutional investors' trading pattern						
Dependent variable = CAR[−1, +1] (%)						
	(1)	Pooled (2)	First court ruling (Event2 = 0)		Second court ruling (Event2 = 1)	
			(3)	(4)	(5)	(6)
N	2,046	2,046	985	985	1,061	1,061
Adj. R ²	3.32%	7.14%	3.54%	10.70%	3.66%	6.44%

Note: This table presents the results of the trading behavior around each court rulings. All the trading behavior variables (i.e., *Institutional investors NB*[−1, +1], *Individual NB*[−1, +1], *Foreign investors NB*[−1, +1], and *Total trading volume*[−1, +1]) are measured from a day before to a day after the announcement date of the court rulings. The variables for net buying of each investors types (i.e., *Institutional investors NB*[−1, +1], *Individual NB*[−1, +1], and *Foreign investors NB*[−1, +1]) are their buying volume minus their selling volume, scaled by the total trading volume of the firm's stock during [−1, +1] period. *Total trading volume*[−1, +1] is the total trading volume of the firm's stock during [−1, +1] period, scaled by the firm's market value in the prior year. Panel A reports the correlation matrix of the trading behavior variables, CARs, and *Treated*. Panels B and C report the results from the OLS regressions where the dependent variables are *Institutional investors NB*[−1, +1] and *Total trading volume*[−1, +1], respectively. Panel D reports the results from the OLS regressions where the dependent variables are CARs. Control variables as in Table 3, Industry fixed effect based on the 2-digit KSIC dummies, and the intercept are included in panels B–D. A detailed explanation of the variables is in Appendix A. PSM denotes the sample matched in Table 5. Robust standard errors adjusted for heteroscedasticity and clustered by firm-level are reported in parentheses.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

examine. If the effect of the second court ruling is stronger than that of the first, then this might suggest a “reinforcing” or “accumulative” effect of a similar precedent. This also supports that the de facto legal binding force of the precedent increases when similar precedents are repeatedly accumulated. If the effect of the first court ruling is stronger than that of the second, then this appears to be close to the circumstance under the common law system. In unreported results, we find that *treated firms* are more likely to experience negative CARs in the second court ruling, compared with that in the first one. Thus, we cautiously suggest a reinforcing effect of the second court ruling that has a similar precedent to the first court ruling about the supermajority provision.

To further support this, we examine whether sophisticated investors sell (or buy) their shares of *treated firms* around the first and second court rulings. Assuming that institutional investors are more likely to be sophisticated in the Korean market (as well as those of other countries), we focus on their trading behavior, particularly the net-buying volume. Furthermore, we note that, if the reinforcing effect is supported, we should observe a more intensive trading behavior of institutional investors on *treated firms* at the second court ruling, compared with those at the first court ruling. We construct the variables for the net-buying volume by investor types similar to the literature on trading behavior (e.g., Barber & Odean, 2008). Specifically, we classify investors in KRX into three types: institutions, individuals, and foreigners.

We first report the correlation matrix for the trading behavior variables, CARs and *Treated* (panel A of Table 6). It shows that CARs and institutional investors' net-buying volume are positively correlated, which is also generally consistent with Barber and Odean (2008). In panel B, we find that institutional investors are more likely to sell their shares of *treated firms* compare with *control firms* around the

second court ruling (columns (5) and (6)), while there is no significant result around the first court ruling (columns (3) and (4)). The results in panel B support that sophisticated investors (e.g., institutions) are aware of the weakened anti-takeover force of *treated firms* and, thus, sell these firms' shares at the second court ruling. Note that this result merely reflects some attention effect for the supermajority provision (typically for *treated firms*). Thus, we examine whether an abnormal trading volume exists for *treated firms* around the court rulings. However, we find no evidence for abnormal trading volume in panel C. Taken as a whole, we argue that institutional investors are sufficiently aware of the two court rulings (or at least the second court ruling) that cast a doubt on the supermajority provision's validity and make it potentially weak.

In panel D of Table 6, we further examine the relationship between CARs and the abovementioned sophisticated investors' net-buying volume. First, at both court rulings, a positive relation between CARs and *Institutional investors NB* exists in general. Second, *Institutional investors NB* × *Treated* are significantly positive around the second court ruling. This indicates a stronger correlation between CARs and *Institutional investors NB* for *treated firms*, compared with *control firms*. Specifically, this result is robust for controlling the firm's total trading volume. In terms of economic significance, column (6) of panel D suggests that for *treated firms*, net-selling by institutional investors which accounts for 10% of the firm's total trading volume, are associated with more negative CARs (about −0.98%), while only about −0.24% for *control firms*.

Overall, based on the results in Table 6, we confirm that the precedent of the court ruling in Korea can actually affect the stock market. Thus, weakening the anti-takeover force of supermajority provision leads to a negative wealth effect, on average, for *treated firms*. In addition, the results in Table 6 can (partially) support the

reinforcing effect of the second court ruling, which has a similar court decision to the first court ruling. Meanwhile, we do not argue that institutional investors' selling strictly drives negative CARs (panel D of Table 6). Rather, we cautiously interpret that our evidence indicates a significant market reaction of *treated firms*, particularly in terms of both negative returns and selling of institutional investors. Our evidence also supports the fact that court rulings were public information in the market, which is in line with our anecdotal evidence in Section 2.2.4.

4.4 | Additional tests for the unchanged and changed sample

The empirical test that we use is based on a quasi-natural experiment design using the two court ruling announcements in line with Cohen and Wang (2013). However, in our study, the interval between the two court rulings in April 2007 and June 2008 is about 14 months, whereas in Cohen and Wang (2013), it is 1 month. Hence, during the 14-month interval between the two court rulings in our study, firms could have revised their charters at their annual meetings; annual meetings are usually held every March in Korea (March 2008 is included in the 14-month interval).²⁴ In other words, our main independent variable could be endogenous for the following two reasons: the market reaction of the first court ruling, which is our dependent

variable, and unobservable (therefore omitted) firm-specific characteristics. In this subsection, to alleviate this potential endogeneity from reverse causality or omitted variable bias, we conduct the following two additional tests.

First, we focus on firms for which anti-takeover provisions were unchanged between the first and second court rulings (hereafter, the unchanged sample). Regardless of whether a firm knows the result of the first court ruling, the anti-takeover provisions of this unchanged sample were decided before the first court ruling and did not subsequently change in the period of analysis. Therefore, this approach is advantageous because such firms (i.e., the observations in the unchanged sample) are less likely to be affected by reverse causality.

Table 7 reports the re-estimation for this unchanged sample. All columns show the coefficients of *Treated* and *Supermajority* are negatively estimated at the 1% (5%) significance level when control variables are excluded (included). Overall, the re-estimations for the unchanged sample are consistent with our main results, showing the causal effect between weakening the anti-takeover force of the supermajority provision and the reduction in shareholders' wealth.

Second, we focus on the sample of firms that changed their anti-takeover provisions (hereafter, the changed sample) to support our main findings. Using this changed sample analysis, we can provide evidence that our results are not driven by reverse causality, and further address the potential endogeneity concern arising from omitted variable bias.

TABLE 7 Unchanged sample analysis

Dependent variable = CAR[−1, +1] (%)						
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	−1.409*** (0.498)	−1.141** (0.508)				
Supermajority			−1.380*** (0.483)	−1.135** (0.491)	−1.363*** (0.494)	−1.110** (0.504)
SB			−0.994 (1.275)	−1.637 (1.507)		
GP			−1.578 (1.045)	−1.249 (1.162)		
Miscell			3.204* (1.726)	3.424* (1.910)		
Supermajority × SB			1.484 (1.462)	1.250 (1.506)		
Supermajority × GP			−1.992 (1.436)	−2.162 (1.811)		
Other-provision					−1.473* (0.756)	−1.469* (0.862)
Supermajority × Other-provision					2.486** (1.169)	2.265* (1.225)
Event2	−0.330 (0.230)	−0.452* (0.234)	−0.393* (0.227)	−0.503** (0.230)	−0.393* (0.226)	−0.504** (0.230)
Control variables	No	Yes	No	Yes	No	Yes
Industry fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
N	1,609	1,590	1,703	1,681	1,703	1,681
Adj. R ²	2.69%	3.91%	2.94%	3.93%	2.95%	4.00%

Note: This table presents the results for the unchanged sample for which the anti-takeover provisions in a firm's charter do not change over the first and second court rulings. Control variables are the same as in Table 3. A detailed explanation of the variables is in Appendix A. Industry fixed effect is based on the 2-digit KSIC dummies, and the intercept is included. Robust standard errors adjusted for heteroscedasticity and clustered by firm-level are reported in parentheses.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

As outlined above, a firm may have decided to change its anti-takeover provision after the first court ruling. For example, a firm with no anti-takeover provisions may have realized that the market views a supermajority provision as a value-enhancing tool (by observing the market reactions to the first court ruling of other firms). Thus, such a firm may have adopted a supermajority provision to maximize its value (i.e., a firm that was a *control firm* at the first court ruling, but a *treated firm* at the second court ruling), even though the first court ruling potentially weakened the anti-takeover force of the supermajority provision. On the other hand, a *treated firm* may eliminate its supermajority provision after the first court ruling (i.e., a *treated firm* at the first court ruling, but a *control firm* at the second court ruling). However, we find no significant relation (untabulated) when we regress the changes of firms' anti-takeover provisions between two court rulings on the returns at the time of the first court ruling announcement.

In Table 8, we conduct both the univariate test and regressions based on the difference-in-differences approach for the changed sample.²⁵ Specifically, we assign a value of one to *Control-to-Treated* for observations that involve *control firms* at the first court ruling but *treated firms* at the second court ruling, and zero for the opposite case. Collectively, we expect that the announcement returns of such firms at the second court ruling will be significantly different from those at the first court ruling. We first compare the CARs of each court ruling for the changed sample (panel A). Consistent with our

prior results, we find that observations for firms that changed from *control firms* to *treated firms* (i.e., *Control-to-Treated* = 1) have negative CARs only at the second court ruling. Observations for firms that changed from *treated firms* to *control firms* (i.e., *Control-to-Treated* = 0) show the opposite direction of CARs between the first and second court rulings. In panel B where we report the results of regressions, all columns show that *control-to-treated firms* have significantly higher announcement returns at the first court ruling, by about 2.13–2.58 percentage points on average, compared with the second court ruling. In other words, *treated firms* still underperform compared with *control firms* at the first court ruling, which is consistent with our previous findings. Most importantly, our main estimator, *Control-to-Treated* × *Event2*, is significantly negative in all columns, suggesting that the market reacts differently (in the opposite direction) at the second court ruling for this changed sample. Specifically, column (3) shows that *control-to-treated firms* underperform on average compared with *treated-to-control firms* at the second court ruling by 2.73 percentage points (= − 5.31 + 2.58). The overall evidence in both panels of Table 8 supports our results.

4.5 | Simulation approach

In this subsection, we use two simulation approaches to support the significance of our main results. First, in panel A of Table 9, we

TABLE 8 Changed sample analysis

Panel A: Univariate test			
	First court ruling (Event2 = 0) (1)	Second court ruling (Event2 = 1) (2)	Difference (3)
CAR[−1, +1]			
Control-to-Treated firms (control-to-treated = 1)	−0.21%	−1.37%*	−1.16%
Treated-to-Control firms (control-to-treated = 0)	−2.67%**	1.14%***	3.81%***
Difference	2.46%*	−2.51%***	
Panel B: Changed sample regression Dependent variable = CAR[−1, +1] (%)			
	(1)	(2)	(3)
Control-to-Treated × Event2	−4.792*** (1.455)	−4.706*** (1.417)	−5.310*** (1.526)
Control-to-Treated	2.131* (1.144)	2.517** (1.079)	2.577** (1.173)
Event2	3.217*** (1.108)	4.001*** (1.095)	3.856*** (1.181)
Control variables	No	Yes	Yes
Industry fixed effect	Yes	No	Yes
N	494	470	470
Adj. R ²	2.57%	3.71%	2.61%

Note: This table presents the results for the changed sample for which the anti-takeover provisions in a firm's charter changed between the first and second court rulings. In panel A, we compare CARs for the changed sample where *Control-to-Treated firms* were *control firms* at the first court ruling but *treated firms* at the second court ruling, and *Treated-to-Control firms* were the opposite cases. Panel B reports the regression results of the changed sample using the difference-in-differences approach. Control variables are the same as in Table 3. A detailed explanation of the variables is in Appendix A. Industry fixed effect is based on the 2-digit KSIC dummies, and the intercept is included. Robust standard errors adjusted for heteroscedasticity and clustered by firm-level are reported in parentheses.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

randomly assign *variable of interest* based on the sample distribution, and report the simulation results, as suggested by Bernile et al. (2017). For example, in column (1), as there were 174 *treated firms* among the 2,060 observations from the original model, we generate a random variable (*Simulated-Treated*) following a binomial distribution, which we set to one with a probability of 0.084 ($= 174/2,060$) and zero with a probability of 0.915. Then, we re-estimate the coefficient of *Simulated-Treated* on the observed announcement returns and record it with the associated *p* value. Similarly, in column (3), we randomly assign the variable, *Control-to-Treated*, in the changed sample, then interact it with the *Event2* variable. We repeat this simulation procedure 1,000 times and randomly re-assign our main independent variable *Simulated-variable* for each replication.

The first row in panel A of Table 9 reports the average coefficient for the randomly re-distributed independent variable (*Simulated-variable*). All columns show that almost zero mean value of the estimated coefficients over the 1,000 repetitions. The second row in panel A reports the percentage of coefficients that are significant at the 5% level in the brackets. All columns show that approximately 4–6% of

the coefficients are significant at the 5% level and that, roughly, those cases are not biased either positively or negatively. Specifically, column (1) shows that the mean estimated coefficients of *Simulated-Treated* for 1,000 simulation trials is 0.011 and that 5.8% of the coefficients are significantly either positive or negative at the 5% level, whereas the estimated coefficient in the original model is -1.520 with a 0.0% *p* value. Moreover, the third row in panel A reports the proportion of replication trials that have a larger negative impact (and significant at the 5% level) than in the original model in parentheses. We find no such trials in columns (1) and (3) and only three trials in column (2).

Second, we compare the effect of the court rulings with a reference distribution that we bootstrap from the non-event days (Cohen & Wang, 2013, 2017). As suggested by Cohen and Wang (2013), this approach assesses how the effect as large as one in the two events can be observed on two random dates (among non-event days). We collect the stock price data of our sample from 100 trading days around event dates for each court ruling. To ensure we avoid the effects of two court rulings clearly, we exclude 15 trading days before and after the rulings; thus, we collect data from non-event trading

TABLE 9 Simulation results

Original model	Pooled (1)	PSM (2)	Changed sample analysis (3)
Variable in interest	Treated	Treated	Control-to-Treated $\times$ Event2
Original β^* for variable in interest	-1.520	-1.788	-5.310
(<i>p</i> value)	(0.0%)	(0.6%)	(0.1%)
Control variables	Yes	Yes	Yes
Industry fixed effect	Yes	Yes	Yes
N	2,060	328	470
Panel A: Random assignment of variable in interest			
Mean β for simulated-variable	0.011	0.004	0.029
[% $\beta > 0$ and $\alpha \leq 5\%$; % $\beta < 0$ and $\alpha \leq 5\%$]	[2.9%; 2.9%]	[2.0%; 1.9%]	[3.2%; 2.3%]
(% $\beta \leq \beta^*$ & $\alpha \leq 5\%$)	(0.0%)	(0.3%)	(0.0%)
Replication trials	1,000	1,000	1,000
Panel B: Market reaction over non-event days			
Mean β for variable in interest	0.258	0.105	-0.053
Simulation <i>p</i> value (% $\beta \leq \beta^*$)	(0.19%)	(2.03%)	(2.18%)

Note: This table presents the statistics of the simulation for the original OLS regression models where the dependent variables are the CARs around the two court rulings. Original β^* indicates that the estimated coefficient of *variable in interest* in the original model and the associated *p* value is shown in parentheses immediately below the original β^* . In panel A, we randomly re-assign the *variable in interest* to each firm based on the sample distribution (*Simulated-variable*) and record the estimated coefficients and corresponding *p* values based on robust standard errors adjusted for heteroscedasticity and clustered by firm-level for each replication. We repeat this simulation procedure 1,000 times. We calculate the mean of the estimated coefficients of *Simulated-variable* across the 1,000 replications. The percentages of the estimated coefficients of *Simulated-variable* that are significantly positive or negative at the 5% level or more are reported in brackets. The percentages of the estimated coefficients of *Simulated-variable* that are significantly negative at the 5% level or more and have a larger absolute value than the original estimates are reported in parentheses. In panel B, we replace our dependent variable with CARs over non-event days and re-estimate the original model. The bottommost rows report the likelihood of observing coefficients at least as large and negative as the original β^* . The simulation *p* value, reported in the bottommost rows, are computed using a simulated null distribution of coefficients generated from all unique pairs of non-event 100 trading days around event days for each court ruling ($[-65, -16]$ and $[+16, +65]$ for events 1 and 2). For example, in column (1) in panel B, the simulation *p* value represents the proportion of times we observe the coefficient of *Treated* for non-event days that is smaller than -1.520 . We also calculate the mean of the estimated coefficients of *variable in interest* above simulation *p* value. Control variables are the same as in Table 3. A detailed explanation of the variables is in Appendix A. Industry fixed effect is based on the 2-digit SIC dummies. PSM denotes the sample matched in Table 5.

days $[-65, -16]$ and $[+16, +65]$ for events 1 and 2. Then, we compute CARs for each 3-day window in the same method as described in Section 3.1. Next, we replace CARs for each of two events and estimate regression models for such CARs over all unique pairs of non-event days.

Panel B of Table 9 reports the results. Column (1) shows that in only 0.19% of our simulation trials, we observe the estimated coefficients of *Treated*—which are from regressing CARs over non-event days on *Treated*—at least as more negative as the observed coefficient in the original model of -1.520 . In column (2), our simulation approach shows that matched *treated firms* underperform compared with matched *control firms* at least about 1.790 percentage points with statistical significance at the 5% level (with a simulation p value of 2.03%). Similar to panel A, we calculate the mean of those estimated coefficients and find an almost zero mean value over our simulation trials. In unreported test, we find consistent results when we use $[-115, -16]$ and $[+16, +115]$ for non-event trading days.

Collectively, the evidence from both panels of Table 9 strongly suggests that our observed market reaction of relatively negative announcement returns for treated firms is unlikely to be the result of random sampling variation or random noise.

4.6 | Robustness tests

In this subsection, we perform several robustness tests. First, for the placebo test, we use a sample of firms with one anti-takeover provision, other than a supermajority provision, as their only anti-takeover provision. The results indicate no statistical differences in the market reaction around the two court rulings between firms with a staggered board or golden parachute as their only anti-takeover provision and *control firms*. Second, we perform the regressions with alternative fixed effects. The results are consistent with our previous estimates. Finally, when we use alternative stock price reaction measures, we find that our previous estimations do not change. For brevity, these results are reported in the Online Appendix.

5 | DISCUSSION

5.1 | Korean corporate governance and the reputation damage hypothesis

In this subsection, we discuss the cross-sectional variation related to Korean corporate governance and the firms' governance quality. First, we note that the corporate governance literature focusing on Korean samples typically emphasizes the large group affiliations (i.e., chaebol firms) and the systematic differences compared with non-chaebol firms (Baek et al., 2004; Baek et al., 2006; Shin & Park, 1999). Therefore, in our previous estimations, it is possible that the market reacts differently to *treated firms* in the chaebol group and those in the non-chaebol group.

In unreported results, we examine the effect of chaebol affiliations using the data from the Korea Fair Trade Commission (KFTC). We first confirm that our main conjecture is not influenced by the small subset of chaebol firms. More interestingly, we find that *treated firms* in the chaebol group are less affected by the two court rulings than are non-chaebol *treated firms*. It seems reasonable that, in general, chaebol firms are strictly controlled by their chaebol group's controlling shareholders and, hence, the effect of the court rulings on their anti-takeover force is less strong (or remains unchanged) compared with non-chaebol *treated firms*.

Second, the market may react more negatively to the lower quality of the firm-level corporate governance (Black et al., 2006), than to *treated firms*. However, we also find that our main results are not driven by an omitted variable bias related to the firm-level corporate governance quality using the firm-level corporate governance quality data from KCGS, as in Lee and Park (2009), and the Korea Economic Justice Institute (KEJI), as in Chang et al. (2017) and Choi et al. (2019).

Finally, we note that our results merely reflect the reputation damage of *treated firms* (hereafter, the reputation damage hypothesis), even though we exclude the respondent firms of the court rulings in our main analysis. Specifically, it is possible that, for *treated firms*, expected reputational losses or high lawsuit risks are reflected in stock prices because they have a higher likelihood of lawsuits similar to those leading the court rulings. However, our previous discussions naturally suggest that this alternative hypothesis is less likely to drive our results. For brevity, we report the untabulated results of this subsection in the Online Appendix and discuss them in detail.

5.2 | Possible channels for the value-enhancing perspective

So far, our empirical results indicate that the market views the supermajority provision as value-enhancing on average, by investigating the relationship between the court rulings weakening the power of the supermajority provision and negative market reactions. Why does the market view such a provision as a tool enhancing shareholders' wealth, rather than a value-destroying one? To answer this question, we discuss possible channels through which the anti-takeover provision (in our study, a supermajority provision) is positively related to firm value in this subsection.

First, the myopic market hypothesis highlights that a firm's long-term projects are undervalued by myopic investors in the market (Chemmanur & Jiao, 2012; Stein, 1988, 1989). Thus, the management of firms with long-term projects may face greater pressure to undertake myopic or short-term investment (Bushee, 1998; Chan et al., 2001; Duru et al., 2013). Second, hard-to-value (HTV) firms are more likely to suffer from a valuation discount due to their higher complexity (Humphery-Jenner, 2014; Momtaz & Drobetz, 2020). Taken as a whole, if our main results support the *value-enhancing perspective* of the supermajority provision, we expect that the subset of firms, such as firms with long-term investment or HTV firms, is more affected by the two court rulings.

TABLE 10 The effect of takeover threat

Dependent variable = CAR[−1, +1] (%)								
Panel A: Takeover threat								
Takeover threat proxy	Relative volume of M&A		Frequency of M&A		Frequency of M&A by domestic acquirers		Frequency of M&A by foreign acquirers	
	Pooled (1)	PSM (2)	Pooled (3)	PSM (4)	Pooled (5)	PSM (6)	Pooled (7)	PSM (8)
Treated	−1.463*** (0.412)	−1.605** (0.665)	−1.146** (0.467)	−1.052 (0.747)	−1.110** (0.463)	−1.040 (0.762)	−1.372*** (0.445)	−1.407** (0.683)
Treated × Takeover threat	−0.064* (0.033)	−0.145** (0.062)	−0.517* (0.309)	−0.939** (0.420)	−0.790* (0.430)	−1.280** (0.543)	−0.723 (0.720)	−1.918* (1.157)
Takeover threat	0.009 (0.031)	0.098 (0.063)	0.218 (0.232)	0.871 (0.751)	0.308 (0.285)	1.502 (0.958)	0.100 (0.815)	−0.265 (2.439)
Event2	−0.041 (0.220)	−1.466** (0.680)	−0.124 (0.246)	−1.615** (0.717)	−0.140 (0.248)	−1.745** (0.718)	−0.044 (0.225)	−1.342* (0.694)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2,060	328	2,060	328	2,060	328	2,060	328
Adj. R ²	2.95%	4.02%	3.05%	4.78%	3.09%	4.97%	2.93%	4.11%
Panel B: Takeover threat and long-term investment effect								
Takeover threat proxy	Relative volume of M&A (1)	Frequency of M&A (2)	Frequency of M&A by domestic acquirers (3)	Frequency of M&A by foreign acquirers (4)	(5)	(6)	(7)	(8)
Treated	−1.550*** (0.421)	−1.499*** (0.573)	−1.174** (0.466)	−1.371** (0.661)	−1.170** (0.465)	−1.286* (0.665)	−1.364*** (0.445)	−1.489** (0.613)
Treated × Takeover threat × R&D expense	−0.044** (0.021)	−0.049** (0.024)	−0.158*** (0.060)	−0.212** (0.092)	−0.193** (0.094)	−0.242* (0.133)	−0.386*** (0.145)	−0.503** (0.240)
Treated × Takeover threat	0.232 (0.154)	0.238 (0.159)	−0.177 (0.392)	−0.077 (0.433)	−0.314 (0.591)	−0.223 (0.650)	0.021 (0.908)	0.197 (0.975)
Takeover threat	0.012 (0.031)	−0.002 (0.035)	0.205 (0.232)	0.171 (0.233)	0.297 (0.284)	0.258 (0.287)	0.062 (0.814)	−0.011 (0.805)
R&D expense (%)	−0.078 (0.055)	−0.081 (0.058)	−0.064 (0.055)	−0.082 (0.061)	−0.067 (0.055)	−0.082 (0.060)	−0.067 (0.055)	−0.077 (0.059)
Treated × R&D expense		−0.024 (0.156)		0.094 (0.184)		0.054 (0.176)		0.063 (0.181)
Takeover threat × R&D expense		0.007 (0.009)		0.027 (0.033)		0.031 (0.039)		0.066 (0.125)
Event2	−0.036 (0.219)	−0.032 (0.219)	−0.112 (0.245)	−0.114 (0.245)	−0.127 (0.248)	−0.128 (0.248)	−0.038 (0.224)	−0.042 (0.224)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2,060	2,060	2,060	2,060	2,060	2,060	2,060	2,060
Adj. R ²	3.04%	2.97%	3.22%	3.16%	3.23%	3.16%	3.07%	2.99%

Note: This table presents the results from the OLS regressions where the dependent variables are the CARs around the two court rulings. In both panels, the four different proxies of *Takeover threat* are as defined in Section 5.2. In panel B, *R&D expense* is the percentage ratio of research and development to total assets. Control variables are the same as in Table 3. A detailed explanation of the variables is in Appendix A. Industry fixed effect is based on the 2-digit KSIC dummies, and the intercept is included. PSM denotes the sample matched in Table 5. Robust standard errors adjusted for heteroscedasticity and clustered by firm-level are reported in parentheses.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

We find partial but supportive evidence that *treated firms* are more likely to underperform when they engage more in long-term projects or belong to HTV industries. These results are reported in the Online Appendix (for brevity). It implies that the supermajority provision as an anti-takeover provision plays a more significant role under certain conditions in our sample, consistent with the *value-enhancing perspective* (e.g., Cremers et al., 2017; Daines & Klausner, 2001). As a result, the supermajority provision can lead firms, typically those with long-term investments or higher complexity, to enhance or at least protect shareholders' wealth.

Finally, we note Chemmanur and Jiao's (2012) argument, that the *value-enhancing perspective* of the anti-takeover provisions is expected to be stronger when the firm faces a high threat of takeover. To examine the effect of takeover threat, we incorporate four proxies as follows: M&A transaction volume relative to the firm's total asset size, M&A frequency, M&A frequency by domestic acquirers, and M&A frequency by foreign acquirers. Similar to Momtaz and Drobetz (2020), we construct these proxies at the within-industry level (the M&A's target industry) over the past 5 years. We obtain M&A data from SDC Platinum.

Table 10 reports the results. All columns in panel A show that *treated firms* underperform more when their industries (as the target industry) have higher M&A volume and frequency. The takeover threat itself does not affect *control firms* around the two court rulings. These results suggest that an intensive takeover threat in the market may cause managerial myopia and can be detrimental to shareholders' wealth. Moreover, if the supermajority provision as a takeover defense ameliorates the threat of opportunistic takeovers and managerial myopia, it can allow managers to implement long-term projects that aim to create firm value. Extending this view, we further examine whether the *treated firms* facing a high takeover threat and with intensive R&D investment in our sample are mostly (relative to other *treated firms*) affected by the two court rulings. In panel B of Table 10, we find that the triple-interaction term of *Treated*, *Takeover threat* and *R&D expense* is significantly negative in all columns, suggesting that the supermajority provision plays a more significant role for firms with a high threat of takeover, particularly when they also have a focus on long-term investment. In contrast to the value-destroying perspective on anti-takeover provisions, the results in Table 10 indicate that takeover threats may not be an effective corporate governance mechanism to promote managers' efforts and to enhance shareholders' wealth.

6 | CONCLUSION

In this paper, we seek to make an empirical contribution to the literature by examining the relationship between the supermajority provision and firm value. As suggested by prior studies that use legal contexts for a quasi-natural experiment (e.g., Cohen & Wang, 2013; Larcker et al., 2011), we design a quasi-natural experiment as our main empirical framework: using court rulings in two major cities in

Korea in 2007 and 2008, which potentially weaken the anti-takeover force of the supermajority provision. This approach enables us to estimate whether participants in the Korean financial market view the supermajority provision as a value-destroying or value-enhancing tool.

Using a sample of non-financial firms listed on the KRX at the time of these two court rulings, we find evidence that is inconsistent with the *value-destroying perspective*, rather supports the *value-enhancing perspective*. First, the average market reaction to the two court rulings is significantly negative for the set of firms most affected (*treated firms*). Second, we find that *treated firms* significantly underperform compared with the non-impacted set of firms using various empirical methodologies, such as PSM and simulation approaches. Third, we find additional evidence that the supermajority provision may have a more significant role for firms with long-term projects, higher complexity, or higher threat of takeover. Overall, our study provides evidence that the market views the supermajority provision as inducing higher shareholders' wealth on average.

One limitation of our study is that we cannot completely rule out the possibility that the effects of the anti-takeover provisions on firm value are heterogeneous among firms or types of anti-takeover provisions. Since our estimates are only the average effect on *treated firms*, thus the supermajority provision might lead to inefficient governance (e.g., managerial entrenchment) in some firms. Given the nature of our empirical design, our finding is also limited to the effect of the supermajority provision, hence we cannot generalize our results to other types of anti-takeover provisions such as a staggered board.

Nevertheless, our finding contributes causal evidence to the long-standing debate on the effect of the anti-takeover provisions on shareholders' wealth. The regulation prohibiting the firm's anti-takeover provisions could be detrimental to shareholders' wealth. Thus, our study finally suggests that policymakers should not unconditionally regard the anti-takeover provisions as value-destroying corporate governance factors and, hence, taking a more contextual view of such provisions is necessary.

ACKNOWLEDGMENTS

We are especially grateful to the editor-in-chief Konstantinos Stathopoulos, the associate editor and two anonymous reviewers for their insightful suggestions. We are also grateful for the helpful comments from Yakov Amihud, Xuan Tian, and seminar participants at the 2019 Conference on Asia-Pacific Financial Markets, the 2019 Australasian Finance and Banking Conference, the 2020 Financial Management Association Annual Meeting, and the 2021 European Financial Management Association Annual Meetings. This paper was previously entitled "Anti-takeover provisions and shareholders wealth: Evidence from a natural experiment on the supermajority provision." Hyeon Joon Kim acknowledges financial support from NRF (National Research Foundation of Korea) Grant funded by the Korean Government (NRF-2018H1A2A1060309-Global Ph.D. Fellowship Program).

CONFLICT OF INTEREST

Hyeong Joon Kim and Seung Hun Han declared no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Hyeong Joon Kim  <https://orcid.org/0000-0001-9179-4974>

Seung Hun Han  <https://orcid.org/0000-0003-2406-9734>

NOTES

- ¹ Consistently, recent studies argue that firms may benefit from the anti-takeover provisions if they have important business relationships with stakeholders (Cen et al., 2016; Johnson et al., 2015).
- ² For example, firms with lower market values may adopt (or continue to maintain) some anti-takeover provisions, which are commonly described as poor governance practices in the literature. In addition, firms can adopt good governance practices for signaling to the market (e.g., signaling that the management will behave well), and thus, the signal may affect stock price, rather than the intrinsic impact of such practices. It is also possible that firms may choose different governance practices endogenously (see, e.g., Adams et al., 2010; Demsetz & Lehn, 1985).
- ³ The Solidarity for Economic Reform (SER) is the non-governmental organization in Korea established in 2006 to protect shareholders right, improve corporate governance, and monitor the government's financial policy.
- ⁴ The report (Economic Reform Report Vol.3 in 2009) is available at the SER website (http://www.eri.or.kr/bbs/board.php?bo_table=B11&wr_id=39&page=11, in Korean, last accessed on July 1, 2019).
- ⁵ Recently, *The Korea Herald*, a media company in Korea, reports in June 2019 that "the supermajority provision for change of directors is the most selected (13%), the golden parachute is the second-most selected (11%), and the supermajority provision for merger is the third-most selected (2%) anti-takeover provision by Korean-listed firms." (<http://biz.heraldcorp.com/view.php?ud=20190625000322>, in Korean, last accessed on July 1, 2019).
- ⁶ In terms of U.S. firms, which are the most commonly examined in the corporate governance literature, the state law determines whether a firm can have a supermajority rule. For example, companies in California cannot adopt any form of supermajority rule if it requires more than 66.67% of the voting rights due to the CA Corp Code § 710 (2016), which states: A supermajority vote is a requirement set forth in the articles or in a certificate of determination authorized under any provision of this division that specified corporate action or actions be approved by a larger proportion of the outstanding shares than a majority, or by a larger proportion of the outstanding shares of a class or series than a majority, but no supermajority vote that is subject to this section shall require a vote in excess of 66 2/3% of the outstanding shares or 66 2/3% of the outstanding shares of any class or series of those shares.
- ⁷ Under the Commercial Act in Korea, matters to be included in the charter are more comprehensive than matters to be included in the U.S. firms' charter. Furthermore, there are a number of differences between Korea and U.S. law and judicial system; Korea is a civil law country, which has German origin, and United States is a common law country, which has English origins (La Porta et al., 1997, 1998, 1999). See Choi et al. (2016) and Choi et al. (2018) for a review of the national characteristics in the Korean judicial system.

- ⁸ For the special-decision in the shareholders' meeting, the Commercial Act in Korea states: "A resolution shall be adopted by the affirmative votes of no less than two thirds of the voting rights of the shareholders present at a general meeting of shareholders and of at least one third of the total issued and outstanding shares (Article § 434 of the Commercial Act in Korea)." The special-decision is required to enforce an important matter, for example, amendments to articles of incorporation, reduction of capital, merger and acquisition, takeover of business, or removal of director. For the general-decision of shareholders meeting, the Commercial Act in Korea states "Unless otherwise provided for by this Act or the articles of incorporation, resolutions shall be adopted at a general meeting of shareholders by affirmative votes of a majority of the voting rights of shareholders present thereat and representing at least a quarter of the total issued and outstanding shares (Article § 368-(1) of the Commercial Act in Korea)."
- ⁹ This news article is available at the website of *Joong-Ang Daily* (<https://news.joins.com/article/2912690>, in Korean, last accessed on July 1, 2019).
- ¹⁰ This news article is available at the website of *The Law Times* (<https://www.lawtimes.co.kr/Legal-News/Legal-News-View?serial=40530>, in Korean, last accessed on July 1, 2019).
- ¹¹ The non-profit organization KCGS has compiled annual information on corporate governance for representative companies on KRX since 2002. For more detail information, we refer the reader to KCGS's website in English: <http://www.cgs.or.kr/eng/index.asp>.
- ¹² This report (KCGS Report Vol.4 No. 8 (April, 2014), p. 2–6) is available at the website of KCGS (http://www.cgs.or.kr/publish/report_view.jsp?tn=53&pp=3&spyear=&skey=&svalue=, in Korean, last accessed on July 1, 2019).
- ¹³ Meanwhile, we could not bring an example for the respondent firm of the first court ruling (FYD Corp.) as KRX had suspended the firm's stock transactions at that time, to protect investors due to the lawsuit from mid-March until mid-May 2007.
- ¹⁴ However, it is possible that the reputation damage for losing the court ruling of respondent firm (GNCO Co., Ltd.) may drive the negative stock price reaction around the court ruling announcement date. Hence, we exclude the respondent firms of the two court rulings in our main analysis, and discuss about the reputation damage hypothesis in detail in Section 5.1 and the Online Appendix. We thank a reviewer for pointing this out.
- ¹⁵ The two court rulings in our study are equivalent to the decision of the U.S. Chancery Court. However, it is worth discussing the fact that two court rulings cannot strictly prohibit the supermajority provision for all firms listed on KRX (for counter example, see CA Corp Code § 710 (2016), which is prohibiting the supermajority vote for all firms located in California). In addition, Korea has the High Court and the Supreme Court, similar to the Court of Appeals and the Supreme Court in the United States, respectively. However, the higher courts in Korea have not yet ruled about the supermajority provision. This point might represent a limitation of our study; nevertheless, we note that our study's underlying assumption is worthwhile to test empirically, in line with Cohen and Wang (2013).
- ¹⁶ Cohen and Wang (2013) multiply the observations associated with the latter court ruling event by negative one because the opposite sign is expected. However, in our empirical setting, we can directly use the returns at the time of announcement of the two events because the expected announcement effect of the two court rulings operate in the same direction.
- ¹⁷ DART is a similar platform to the Electronic Data Gathering, Analysis, and Retrieval (EDGAR) in the United States. The firms listed on the Korean market should announce their annual report on DART every fiscal year (and any major firm changes to protect the investors). DART also requires a restricted format for annual report similar to the

U.S. SEC's Form 10-K, including a financial statement, an income statement, and a firm's charter.

- ¹⁸ We consider announcements that have potential effect on the stock return as follows: initial public offerings (IPOs), delisting, mergers and acquisitions, split-up, CEO turnover, capital reduction, bonus issues, major change of the business purpose or company identity, investors relation (IR) presentation, litigation-related announcement, capital expenditure, stock split, stock repurchase, business suspension, seasoned equity offerings (SEOs), convertible bonds and bonds with warrants offerings, corporate bond offerings, sales of fixed assets in the firm, dividend of stock or cash, and patent application.
- ¹⁹ In unreported tests, we exclude observations with such events around 3 days, a week, or a month before and after the dates of two court rulings (rather than the 2 weeks of the main analysis), and re-estimate the baseline regression. The estimated coefficients of *Treated* remain significant and negative at the 1% level. Overall, we find that our main results are not largely changed in these unreported tests.
- ²⁰ There are several provisions for takeover defenses in the governance index measures for U.S. firms (e.g., FK-index of Field & Karpoff, 2002, G-index of Gompers et al., 2003, and E-index of Bebchuk et al., 2009). However, owing to differences in commercial law between the United States and Korea, we only consider the three types of takeover defenses (as we describe in Section 2.1): *Supermajority*, *Staggered Board*, and *Golden Parachute*. For example, *Poison Pill* and *Unequal Voting Rights* are not legally allowed in Korea. We manually check for all the possible anti-takeover provisions during our data-collection process. Thus, we identify *Miscellaneous provisions* such as “board can be removed only for cause,” “only persons who have been in the company for more than 3 years can be appointed as directors” and so on.
- ²¹ Note that our estimates are the average treatment effect on treated (ATET), whereas Cohen and Wang (2013, 2017) argue that their estimates are the average treatment effect (ATE). Therefore, we perform various empirical specifications and discuss the results in later sections to address potential endogeneity concerns. We thank a reviewer for pointing this out.
- ²² Meanwhile, *control firms* show the positive stock price reaction on average. These results may raise some concern about measurement error. However, we emphasize that the main focus of this study is not on testing the effectiveness (or validation) of the market model in our sample. If we assume that the market model overall overestimates abnormal returns in our sample, then the actual effect of two court rulings on *treated firms* would be more negative than our estimates. In sum, we argue that regardless of whether the market model on April 13, 2007, and June 2, 2008, was sufficiently efficient, the difference between *treated* and *control firms* is crucial for our study. In addition, we find consistent results overall when we use some alternative measures of stock price reaction in the Online Appendix.
- ²³ Our matching results are largely unchanged when we add the industry (the first-two KSIC digits) in the matching criteria. However, due to the limited sample size, this result is untabulated.
- ²⁴ In general, many firms on the KRX have a fiscal year ending at the end of December and hold their annual meeting in March. Therefore, the last Friday (or last week) of March is usually called “Super-day (or Super-week) for annual meetings” in Korea. For example, *Joong-Ang Daily*, the major media company in Korea reports “Approximately 26% and 50% of firms listed on KRX hold shareholders meeting during the Super-day and Super-week for annual-meeting, respectively” (<https://news.joins.com/article/22440615>, in Korean, last accessed on July 1, 2019).
- ²⁵ Note that the firm's decision to change its anti-takeover provisions could be endogenous even the court rulings per se are explicitly exogenous to each firm's such decision. We thank a reviewer for pointing this out. Therefore, we cautiously interpret the results of the difference-in-differences approach for this changed sample analysis. We emphasize

that the wealth effect of the second court ruling for this changed sample is significantly different from that of the first court ruling through our results in this difference-in-differences regression (panel B of Table 8).

REFERENCES

- Adams, R. B., Hermalin, B. E., & Weisbach, M. S. (2010). The role of boards of directors in corporate governance: A conceptual framework and survey. *Journal of Economic Literature*, 48, 58–107. <https://doi.org/10.1257/jel.48.1.58>
- Amihud, Y., Schmid, M., & Davidoff Solomon, S. (2017). Do staggered boards affect firm value? Unpublished working paper.
- Amihud, Y., & Stoyanov, S. (2017). Do staggered boards harm shareholders? *Journal of Financial Economics*, 123, 432–439. <https://doi.org/10.1016/j.jfineco.2016.04.002>
- Baek, J. S., Kang, J. K., & Lee, I. (2006). Business groups and tunneling: Evidence from private securities offerings by Korean chaebols. *The Journal of Finance*, 61, 2415–2449. <https://doi.org/10.1111/j.1540-6261.2006.01062.x>
- Baek, J.-S., Kang, J.-K., & Park, K. S. (2004). Corporate governance and firm value: Evidence from the Korean financial crisis. *Journal of Financial Economics*, 71, 265–313. [https://doi.org/10.1016/S0304-405X\(03\)00167-3](https://doi.org/10.1016/S0304-405X(03)00167-3)
- Barber, B. M., & Odean, T. (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *The Review of Financial Studies*, 21, 785–818. <https://doi.org/10.1093/rfs/hhm079>
- Bebchuk, L., Cohen, A., & Ferrell, A. (2009). What matters in corporate governance? *The Review of Financial Studies*, 22, 783–827. <https://doi.org/10.1093/rfs/hhn099>
- Bebchuk, L. A., Coates, J. C. IV, & Subramanian, G. (2002). The powerful antitakeover force of staggered boards: Further findings and a reply to symposium participants. *Stan. L. Rev.*, 55(3), 885–917. <https://doi.org/10.2307/1229675>
- Bebchuk, L. A., & Cohen, A. (2005). The costs of entrenched boards. *Journal of Financial Economics*, 78, 409–433. <https://doi.org/10.1016/j.jfineco.2004.12.006>
- Bernile, G., Bhagwat, V., & Rau, P. R. (2017). What doesn't kill you will only make you more risk-loving: Early-life disasters and CEO behavior. *The Journal of Finance*, 72, 167–206. <https://doi.org/10.1111/jofi.12432>
- Bertrand, M., & Mullainathan, S. (2003). Enjoying the quiet life? Corporate governance and managerial preferences. *Journal of Political Economy*, 111, 1043–1075. <https://doi.org/10.1086/376950>
- Bhargava, R., Faircloth, S., & Zeng, H. (2017). Takeover protection and stock price crash risk: Evidence from state antitakeover laws. *Journal of Business Research*, 70, 177–184. <https://doi.org/10.1016/j.jbusres.2016.08.021>
- Bhojraj, S., Sengupta, P., & Zhang, S. (2017). Takeover defenses: Entrenchment and efficiency. *Journal of Accounting and Economics*, 63, 142–160. <https://doi.org/10.1016/j.jaccoco.2016.12.001>
- Black, B. S., Jang, H., & Kim, W. (2006). Does corporate governance predict firms' market values? Evidence from Korea. *The Journal of Law, Economics, and Organization*, 22, 366–413. <https://doi.org/10.1093/jleo/ewj018>
- Bushee, B. J. (1998). The influence of institutional investors on myopic R&D investment behavior. *Accounting Review*, 305–333. <https://www.jstor.org/stable/248542>
- Cain, M. D., McKeon, S. B., & Solomon, S. D. (2017). Do takeover laws matter? Evidence from five decades of hostile takeovers. *Journal of Financial Economics*, 124, 464–485. <https://doi.org/10.1016/j.jfineco.2017.04.003>
- Cen, L., Dasgupta, S., & Sen, R. (2016). Discipline or disruption? Stakeholder relationships and the effect of takeover threat. *Management Science*, 62, 2820–2841. <https://doi.org/10.1287/mnsc.2015.2252>

- Chan, L. K., Lakonishok, J., & Sougiannis, T. (2001). The stock market valuation of research and development expenditures. *The Journal of Finance*, 56, 2431–2456. <https://doi.org/10.1111/0022-1082.00411>
- Chang, Y. K., Oh, W.-Y., Park, J. H., & Jang, M. G. (2017). Exploring the relationship between board characteristics and CSR: Empirical evidence from Korea. *Journal of Business Ethics*, 140, 225–242. <https://doi.org/10.1007/s10551-015-2651-z>
- Chemmanur, T. J., & Jiao, Y. (2012). Dual class IPOs: A theoretical analysis. *Journal of Banking & Finance*, 36, 305–319. <https://doi.org/10.1016/j.jbankfin.2011.07.010>
- Choi, H., Kang, H. G., Kim, W., Lee, C., & Park, J. (2016). Too big to jail? Company status and judicial bias in an emerging market. *Corporate Governance: An International Review*, 24, 85–104. <https://doi.org/10.1111/corg.12142>
- Choi, H., Kang, H. G., & Lee, C. (2018). What constitutes “too big to jail”? Evidence from South Korea's family business groups. *Asia-Pacific Journal of Financial Studies*, 47, 881–919. <https://doi.org/10.1111/ajfs.12241>
- Choi, Y. K., Han, S. H., & Kwon, Y. (2019). CSR activities and internal capital markets: Evidence from Korean business groups. *Pacific-Basin Finance Journal*, 55, 283–298. <https://doi.org/10.1016/j.pacfin.2019.04.008>
- Cohen, A., & Wang, C. C. (2013). How do staggered boards affect shareholder value? Evidence from a natural experiment. *Journal of Financial Economics*, 110, 627–641. <https://doi.org/10.1016/j.jfineco.2013.08.005>
- Cohen, A., & Wang, C. C. (2017). Reexamining staggered boards and shareholder value. *Journal of Financial Economics*, 125, 637–647. <https://doi.org/10.1016/j.jfineco.2017.06.004>
- Cremers, K. M., Litov, L. P., & Sepe, S. M. (2017). Staggered boards and long-term firm value, revisited. *Journal of Financial Economics*, 126, 422–444. <https://doi.org/10.1016/j.jfineco.2017.08.003>
- Cremers, M., & Ferrell, A. (2014). Thirty years of shareholder rights and firm value. *The Journal of Finance*, 69, 1167–1196. <https://doi.org/10.1111/jofi.12138>
- Daines, R. (2001). Does Delaware law improve firm value? *Journal of Financial Economics*, 62, 525–558. [https://doi.org/10.1016/S0304-405X\(01\)00086-1](https://doi.org/10.1016/S0304-405X(01)00086-1)
- Daines, R., & Klausner, M. (2001). Do IPO charters maximize firm value? Antitakeover protection in IPOs. *Journal of Law, Economics, and Organization*, 17, 83–120. <https://doi.org/10.1093/jleo/17.1.83>
- Daines, R., Li, S. X., & Wang, C. C. (2018). Can staggered boards improve value? Evidence from the Massachusetts natural experiment. Unpublished working paper
- DeAngelo, H., & Rice, E. M. (1983). Antitakeover charter amendments and stockholder wealth. *Journal of Financial Economics*, 11, 329–359. [https://doi.org/10.1016/0304-405X\(83\)90016-8](https://doi.org/10.1016/0304-405X(83)90016-8)
- Demsetz, H., & Lehn, K. (1985). The structure of corporate ownership: Causes and consequences. *Journal of Political Economy*, 93, 1155–1177. <https://doi.org/10.1086/261354>
- Duru, A., Wang, D., & Zhao, Y. (2013). Staggered boards, corporate opacity and firm value. *Journal of Banking & Finance*, 37, 341–360. <https://doi.org/10.1016/j.jbankfin.2012.09.002>
- Faleye, O. (2007). Classified boards, firm value, and managerial entrenchment. *Journal of Financial Economics*, 83, 501–529. <https://doi.org/10.1016/j.jfineco.2006.01.005>
- Field, L. C., & Karpoff, J. M. (2002). Takeover defenses of IPO firms. *The Journal of Finance*, 57, 1857–1889. <https://doi.org/10.1111/0022-1082.00482>
- Frattaroli, M. (2020). Does protectionist anti-takeover legislation lead to managerial entrenchment? *Journal of Financial Economics*, 136, 106–136. <https://doi.org/10.1016/j.jfineco.2019.03.014>
- Giroud, X., & Mueller, H. M. (2011). Corporate governance, product market competition, and equity prices. *The Journal of Finance*, 66, 563–600. <https://doi.org/10.1111/j.1540-6261.2010.01642.x>
- Gompers, P., Ishii, J., & Metrick, A. (2003). Corporate governance and equity prices. *The Quarterly Journal of Economics*, 118, 107–156. <https://doi.org/10.1162/00335530360535162>
- Graham, J. R., Michaely, R., & Roberts, M. R. (2003). Do price discreteness and transactions costs affect stock returns? Comparing ex-dividend pricing before and after decimalization. *The Journal of Finance*, 58, 2611–2636. <https://doi.org/10.1046/j.1540-6261.2003.00617.x>
- Harris, M., & Raviv, A. (1988). Corporate governance: Voting rights and majority rules. *Journal of Financial Economics*, 20, 203–235. [https://doi.org/10.1016/0304-405X\(88\)90045-1](https://doi.org/10.1016/0304-405X(88)90045-1)
- Heckman, J. J., Ichimura, H., & Todd, P. (1998). Matching as an econometric evaluation estimator. *The Review of Economic Studies*, 65, 261–294. <https://doi.org/10.1111/1467-937X.00044>
- Heckman, J. J., Ichimura, H., & Todd, P. E. (1997). Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *The Review of Economic Studies*, 64, 605–654. <https://doi.org/10.2307/2971733>
- Humphery-Jenner, M. (2014). Takeover defenses, innovation, and value creation: Evidence from acquisition decisions. *Strategic Management Journal*, 35, 668–690. <https://doi.org/10.1002/smj.2121>
- Jensen, M. C. (1993). The modern industrial revolution, exit, and the failure of internal control systems. *The Journal of Finance*, 48, 831–880. <https://doi.org/10.1111/j.1540-6261.1993.tb04022.x>
- Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3, 305–360. [https://doi.org/10.1016/0304-405X\(76\)90026-X](https://doi.org/10.1016/0304-405X(76)90026-X)
- Johnson, W. C., Karpoff, J. M., & Yi, S. (2015). The bonding hypothesis of takeover defenses: Evidence from IPO firms. *Journal of Financial Economics*, 117, 307–332. <https://doi.org/10.1016/j.jfineco.2015.03.008>
- Karpoff, J. M., & Malatesta, P. H. (1989). The wealth effects of second-generation state takeover legislation. *Journal of Financial Economics*, 25, 291–322. [https://doi.org/10.1016/0304-405X\(89\)90085-8](https://doi.org/10.1016/0304-405X(89)90085-8)
- Karpoff, J. M., & Wittry, M. D. (2018). Institutional and legal context in natural experiments: The case of state antitakeover laws. *The Journal of Finance*, 73, 657–714. <https://doi.org/10.1111/jofi.12600>
- Knoeber, C. R. (1986). Golden parachutes, shark repellents, and hostile tender offers. *The American Economic Review*, 76, 155–167. <https://www.jstor.org/stable/1804133>
- La Porta, R., Lopez-de-Silanes, F., Shleifer, A., & Vishny, R. W. (1997). Legal determinants of external finance. *The Journal of Finance*, 52, 1131–1150. <https://doi.org/10.1111/j.1540-6261.1997.tb02727.x>
- La Porta, R., Lopez-de-Silanes, F., Shleifer, A., & Vishny, R. W. (1998). Law and finance. *Journal of Political Economy*, 106, 1113–1155. <https://doi.org/10.1086/250042>
- La Porta, R., Lopez-de-Silanes, F., Shleifer, A., & Vishny, R. W. (1999). The quality of government. *Journal of Law, Economics, and Organization*, 15, 222–279. <https://doi.org/10.1093/jleo/15.1.222>
- Larcker, D. F., Ormazabal, G., & Taylor, D. J. (2011). The market reaction to corporate governance regulation. *Journal of Financial Economics*, 101, 431–448. <https://doi.org/10.1016/j.jfineco.2011.03.002>
- Lee, D. W., & Park, K. S. (2009). Does institutional activism increase shareholder wealth? Evidence from spillovers on non-target companies. *Journal of Corporate Finance*, 15, 488–504. <https://doi.org/10.1016/j.jcorpfin.2009.03.002>
- Manne, H. G. (1965). Mergers and the market for corporate control. *Journal of Political Economy*, 73, 110–120. <https://doi.org/10.1086/259000>
- Martijn Cremers, K., Nair, V. B., & John, K. (2009). Takeovers and the cross-section of returns. *Review of Financial Studies*, 22(4), 1409–1445. <https://doi.org/10.1093/rfs/hhn032>
- Masulis, R. W., Wang, C., & Xie, F. (2007). Corporate governance and acquirer returns. *The Journal of Finance*, 62, 1851–1889. <https://doi.org/10.1111/j.1540-6261.2007.01259.x>

- Momtaz, P., & Drobetz, W. (2020). Antitakeover provisions and firm value: New evidence from the M&A market. *Journal of Corporate Finance*, 62, 101594. <https://doi.org/10.1016/j.jcorpfin.2020.101594>
- Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70, 41–55. <https://doi.org/10.1093/biomet/70.1.41>
- Shin, H.-H., & Park, Y. S. (1999). Financing constraints and internal capital markets: Evidence from Korean 'chaebols'. *Journal of Corporate Finance*, 5, 169–191. [https://doi.org/10.1016/S0929-1199\(99\)00002-4](https://doi.org/10.1016/S0929-1199(99)00002-4)
- Shleifer, A., & Summers, L. H. (1988). Breach of trust in hostile takeovers. In A. J. Auerbach (Ed.), *Corporate takeovers: Causes and consequences* (pp. 33–68). University of Chicago Press.
- Smith, J. A., & Todd, P. E. (2005). Does matching overcome LaLonde's critique of nonexperimental estimators? *Journal of Econometrics*, 125, 305–353. <https://doi.org/10.1016/j.jeconom.2004.04.011>
- Stein, J. C. (1988). Takeover threats and managerial myopia. *Journal of Political Economy*, 96, 61–80. <https://doi.org/10.1086/261524>
- Stein, J. C. (1989). Efficient capital markets, inefficient firms: A model of myopic corporate behavior. *The Quarterly Journal of Economics*, 104, 655–669. <https://doi.org/10.2307/2937861>
- Stulz, R. (1988). Managerial control of voting rights: Financing policies and the market for corporate control. *Journal of Financial Economics*, 20, 25–54. [https://doi.org/10.1016/0304-405X\(88\)90039-6](https://doi.org/10.1016/0304-405X(88)90039-6)

AUTHOR BIOGRAPHIES

Hyeon Joon Kim is a visiting professor at the Dongguk Business School, Dongguk University, Republic of Korea. He received his

Ph.D. degree from the Korea Advanced Institute of Science and Technology (KAIST), and this paper is based on his Ph.D. dissertation at KAIST. His research interests include corporate governance and CEO characteristics.

Seung Hun Han is a professor of finance at the School of Business and Technology Management, KAIST, Republic of Korea. He received his Ph.D. in finance from the University of Central Florida. His research interest includes corporate governance, corporate environmental management, mergers and acquisitions, and bond market analysis.

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Kim, H. J., & Han, S. H. (2022). Supermajority provisions and shareholders wealth: Evidence from South Korea's natural experiment. *Corporate Governance: An International Review*, 30(3), 311–334. <https://doi.org/10.1111/corg.12402>

APPENDIX A: DEFINITION OF THE VARIABLES

Variable name	Definition/measurements	Data source
CAR[−1, +1] (%)	The cumulative abnormal return (CAR) as percentage measured over the window [−1, +1] relative to the two court ruling announcement dates, calculated using the market model estimated over the past 180 trading days; the first court ruling date was April 13, 2007, and the second one was June 2, 2008.	FN Dataguide
Supermajority	1 if the firm has the supermajority voting rule (to approve the merger, change directors, and change charter provisions) in its charter, or if the firm has minimum attendance requirement for the shareholders' meeting; 0 otherwise.	DART
Staggered Board (SB)	1 if the firm has a staggered board structure in its charter; 0 otherwise.	DART
Golden Parachute (GP)	1 if the firm has a golden parachute rule in its charter; 0 otherwise.	DART
Miscellaneous provisions (Miscell)	1 if the firm has a miscellaneous anti-takeover provision (e.g., only people who have been in the company for more than 3 years can be appointed as directors) in its charter; 0 otherwise.	DART
Other-provision	1 if the firm has at least one of the following: a staggered board structure, a golden parachute rule, and a miscellaneous provision in its charter (i.e., Other-provision = 1 if SB = 1 or GP = 1 or Miscell = 1); 0 otherwise.	DART
Treated	1 if the firm has the supermajority rule as the only anti-takeover provision in its charter (<i>treated firms</i>); 0 if the firm has no anti-takeover provision (<i>control firms</i>).	DART
Event2	1 for the sample of the second court ruling; 0 for the sample of the first court ruling.	FN Dataguide
Industry dummies	The first two-digit Korean Standard Industrial Classification (KSIC) code dummies (Industry Fixed Effect), where KSIC is provided by Statistics Korea, a central organization for statistics under the Ministry of Strategy and Finance.	FN Dataguide
Age	The fiscal year prior to the announcement minus the year of corporate foundation plus one	FN Dataguide

(Continues)

Variable name	Definition/measurements	Data source
Size	Natural logarithm of total assets (thousands of Korean won [KRW]) in the fiscal year prior to the announcement.	FN Dataguide
Cash	Cash and cash equivalents divided by total assets in the fiscal year prior to the announcement.	FN Dataguide
Leverage	The book value of liabilities divided by total assets in the fiscal year prior to the announcement.	FN Dataguide
M-to-B	Market-to-book value; total market capitalization divided by the book value of equity in the fiscal year prior to the announcement.	FN Dataguide
OCF	Operating cash flows divided by total assets in the fiscal year prior to the announcement.	FN Dataguide
ROA	Net income divided by total assets in the fiscal year prior to the announcement.	FN Dataguide
Investment	The total expenses of research and development and capital expenditure divided by total assets in the fiscal year prior to the announcement.	FN Dataguide
Stock volatility	The cumulative standard deviation of stock prices for a month (about 20 trading days) at the date 2 weeks before the court ruling announcements.	FN Dataguide
Stock run-up	Cumulative stock return for a month (about 20 trading days) at the date 2 weeks before the court ruling announcements.	FN Dataguide
Board size	Natural logarithm of the number of directors in the fiscal year prior to the announcement.	TS2000
Board independence	The ratio of the number of outside directors to the number of the directors in the fiscal year prior to the announcement.	TS2000
Controlling shareholder ownership	Proportion of shares owned by the controlling shareholder of the firm in the fiscal year prior to the announcement.	FN Dataguide
Institutional investors NB [−1, +1] (%)	Institutional investors (including private equity funds and the pension funds)' net-buying (NB) of the firm as percentage measured over the window [−1, +1] relative to the two court ruling announcement dates, where NB is the buying KRW volume minus the selling KRW volume, scaled by total trading KRW volume.	FN Dataguide
Individual investors NB [−1, +1] (%)	Individual investors' net-buying (NB) of the firm as percentage measured over the window [−1, +1] relative to the two court ruling announcement dates, where NB is the buying KRW volume minus the selling KRW volume, scaled by total trading KRW volume.	FN Dataguide
Foreign investors NB[−1, +1] (%)	Foreign investors' net-buying (NB) of the firm as percentage measured over the window [−1, +1] relative to the two court ruling announcement dates, where NB is the buying KRW volume minus the selling KRW volume, scaled by total trading KRW volume.	FN Dataguide
Total trading volume[−1, +1] (%)	Total trading volume (percentage expressed) of the firm over the window [−1, +1] relative to the two court ruling announcement dates, scaled by the firm's market value in the fiscal year prior to the announcement.	FN Dataguide
Control-to-Treated	1 if the firm was a <i>control firm</i> at the first court ruling but a <i>treated firm</i> at the second court ruling, or a <i>treated firm</i> at the second court ruling but no observation for the first court ruling, or a <i>control firm</i> at the first court ruling but no observation for the second court ruling; 0 for vice-versa.	DART
Relative volume of M&A (Takeover threat)	The ratio of total transaction volume of mergers and acquisitions (M&A) to the firm's total assets, where M&A are defined as those in the past 5 years targeting the industry to which the firm belongs.	SDC Platinum Mergers and Acquisitions
Frequency of M&A (Takeover threat)	The total number of mergers and acquisitions (M&A), over the past 5 years targeting the industry to which the firm belongs.	SDC Platinum Mergers and Acquisitions
Frequency of M&A by domestic acquirers (Takeover threat)	The total number of mergers and acquisitions (M&A) by domestic acquirers (Korean firms), over the past 5 years targeting the industry to which the firm belongs.	SDC Platinum Mergers and Acquisitions
Frequency of M&A by foreign acquirers (Takeover threat)	The total number of mergers and acquisitions (M&A) by foreign acquirer (non-Korean firms), over the past 5 years targeting the industry to which the firm belongs.	SDC Platinum Mergers and Acquisitions